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Li et al.

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(54) **PLUG CONNECTOR AND CONNECTOR ASSEMBLY CAPABLE OF ACHIEVING MINIATURIZATION**

(71) Applicant: **DONGGUAN LUXSHARE TECHNOLOGIES CO., LTD,**
Dongguan (CN)

(72) Inventors: **Tiesheng Li**, Dongguan (CN); **Rongzhe Guo**, Dongguan (CN); **Hongji Chen**, Dongguan (CN); **Zhaolong Zeng**, Dongguan (CN); **Sihai Yi**, Dongguan (CN)

(73) Assignee: **DONGGUAN LUXSHARE TECHNOLOGIES CO., LTD,**
Dongguan (CN)

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H01R 12/71 (2011.01)
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(Continued)

(58) **Field of Classification Search**
CPC H01R 13/6471; H01R 13/516; H01R 13/629; H01R 13/665; H01R 12/71
USPC 439/108
See application file for complete search history.

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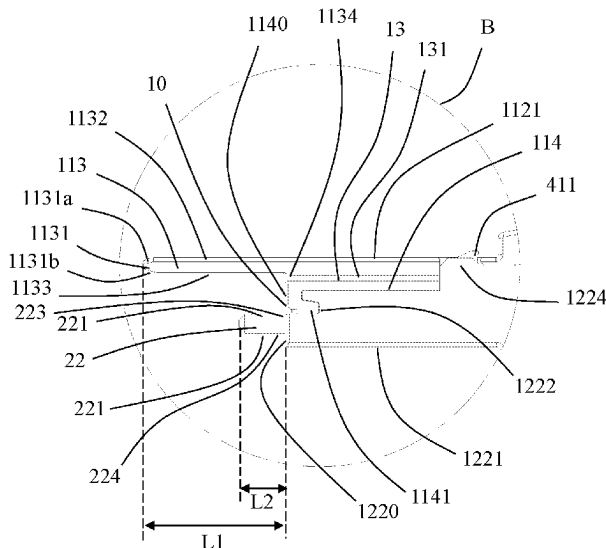
Primary Examiner — Gary F Paumen

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch & Birch, LLP

(57) **ABSTRACT**

A connector assembly includes a receptacle connector assembly and a plug connector. The plug connector includes a housing and a tongue plate. The housing includes a mating surface. A number of conductive pads are provided on the tongue plate. The housing includes an extension plate extending in a mating direction of the plug connector. The tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction. The plug connector is provided with the extension plate which is located only on one side of the tongue plate. The extension plate extends and protrudes beyond the tongue plate along the mating direction.

20 Claims, 30 Drawing Sheets



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H01R 13/629 (2006.01)
H01R 13/66 (2006.01)
H01R 13/627 (2006.01)
H01R 13/633 (2006.01)

(52) **U.S. Cl.**

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 (2013.01); **H01R 13/6275** (2013.01); **H01R**
13/6335 (2013.01)

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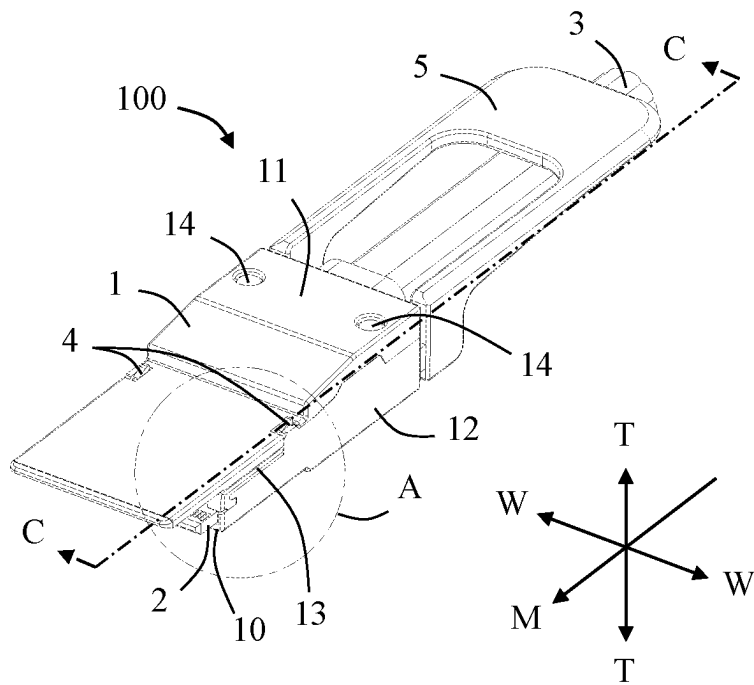


FIG. 1

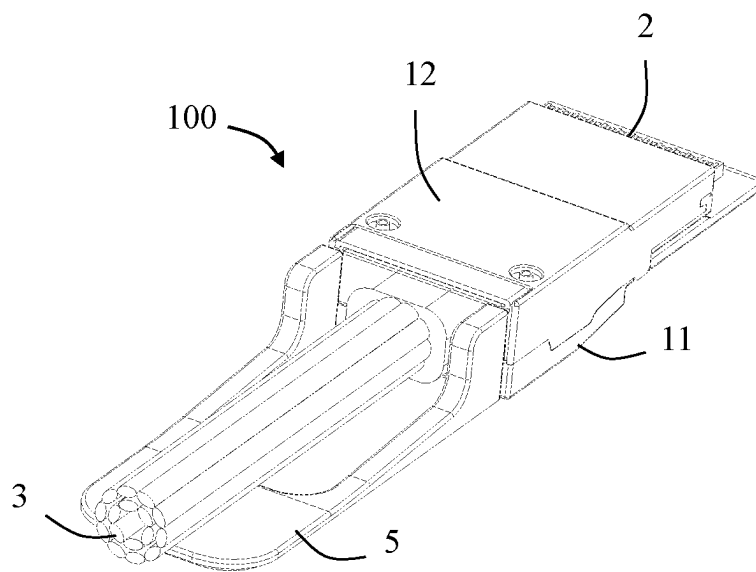


FIG. 2

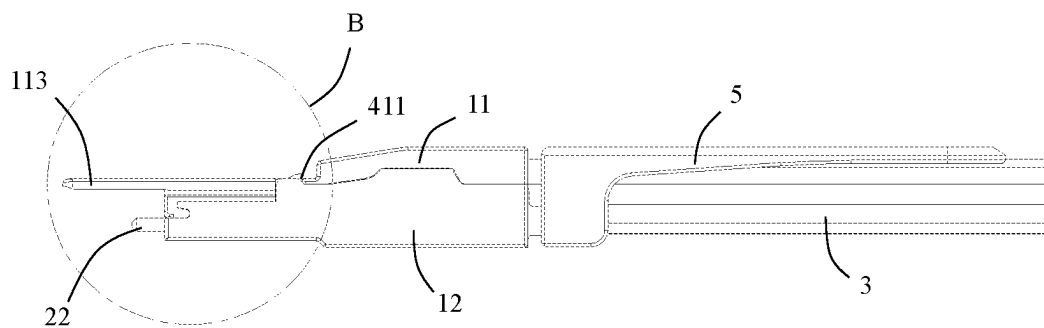


FIG. 3

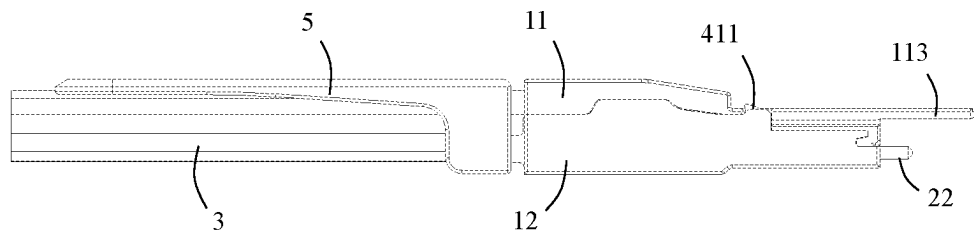


FIG. 4

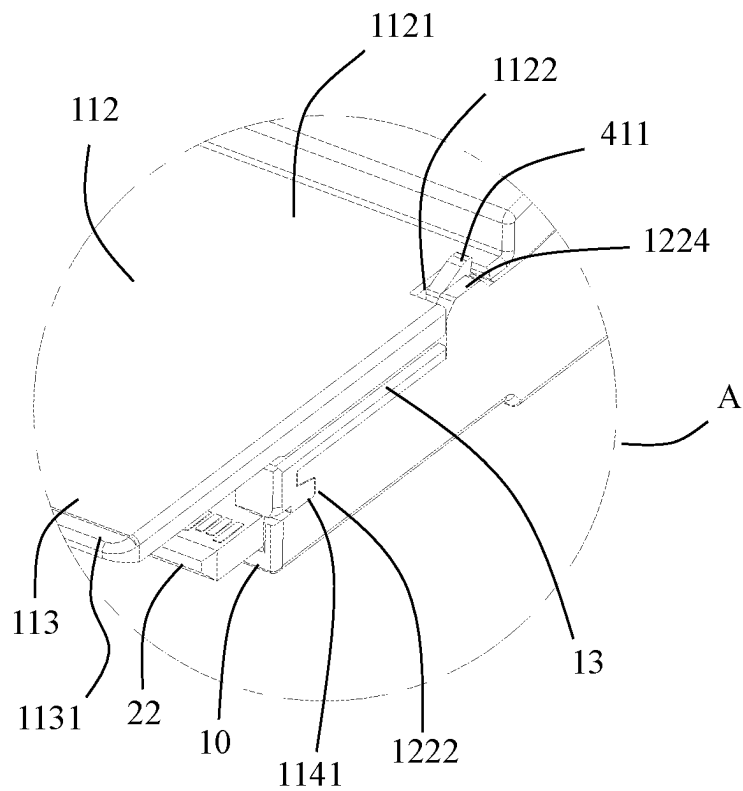


FIG. 5

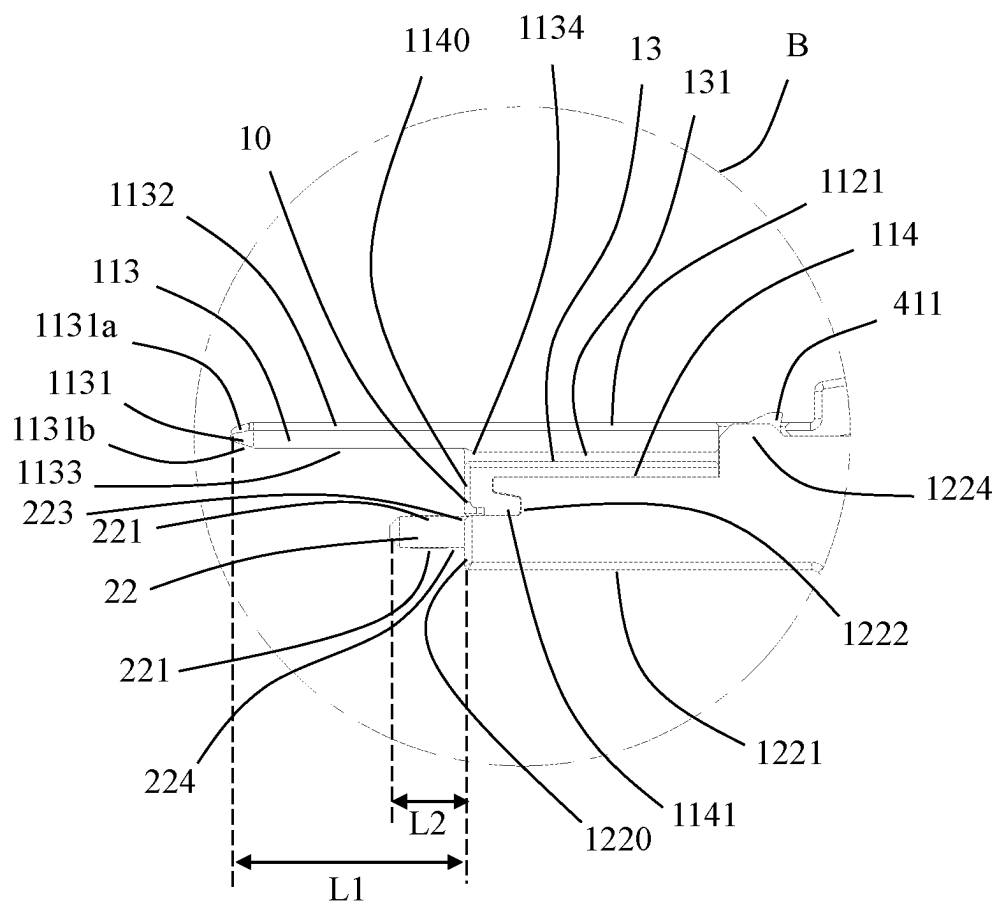


FIG. 6

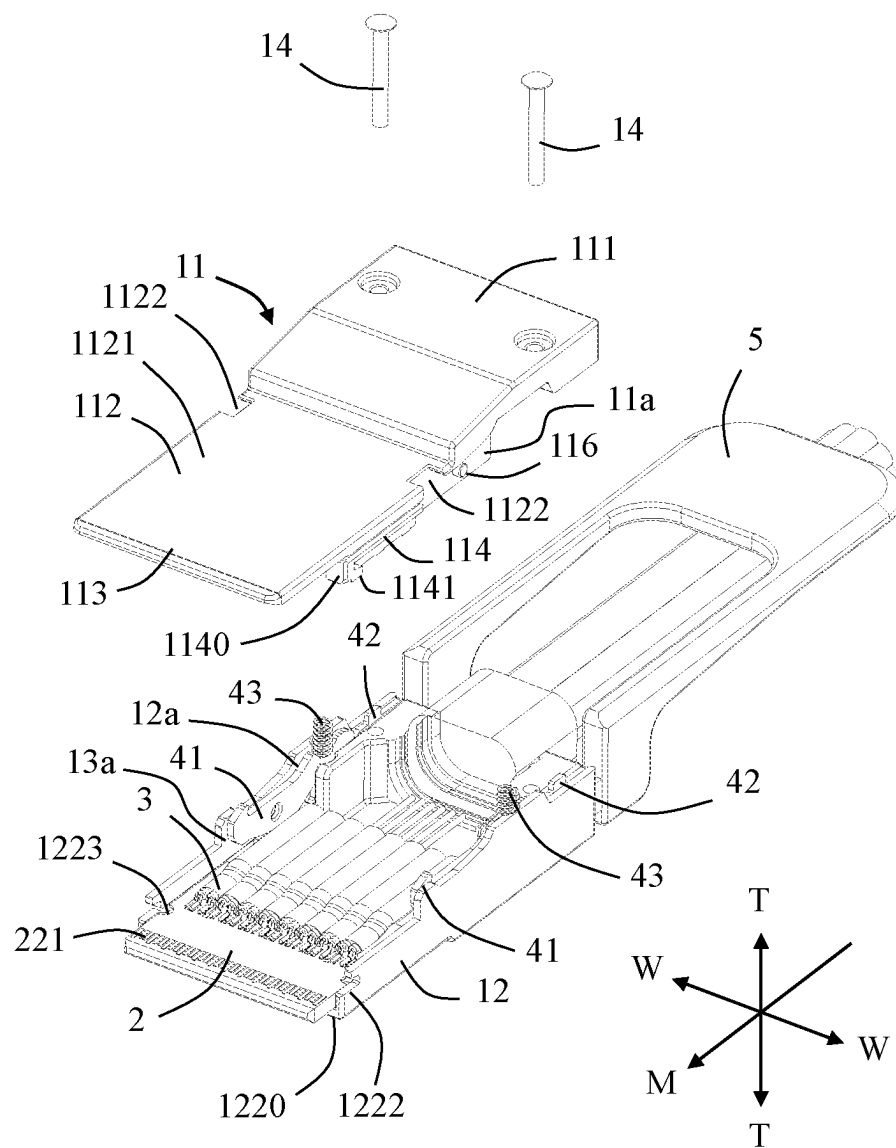


FIG. 7

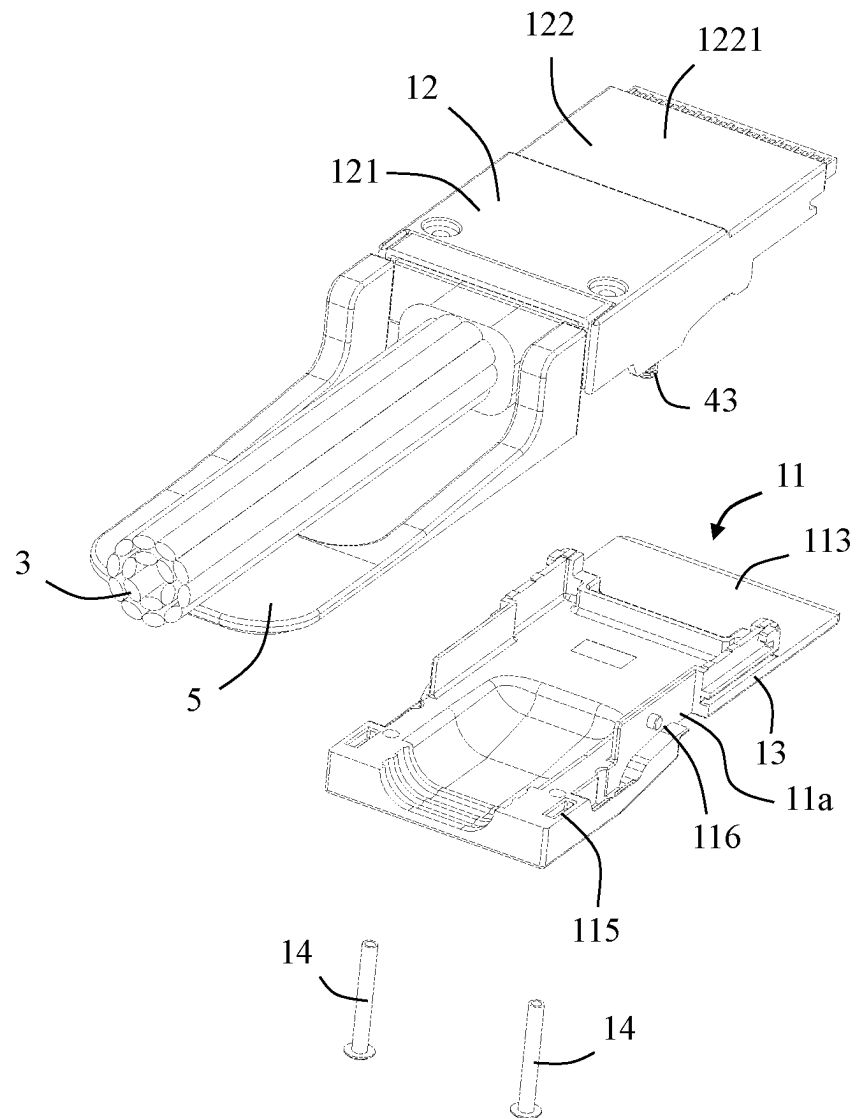


FIG. 8

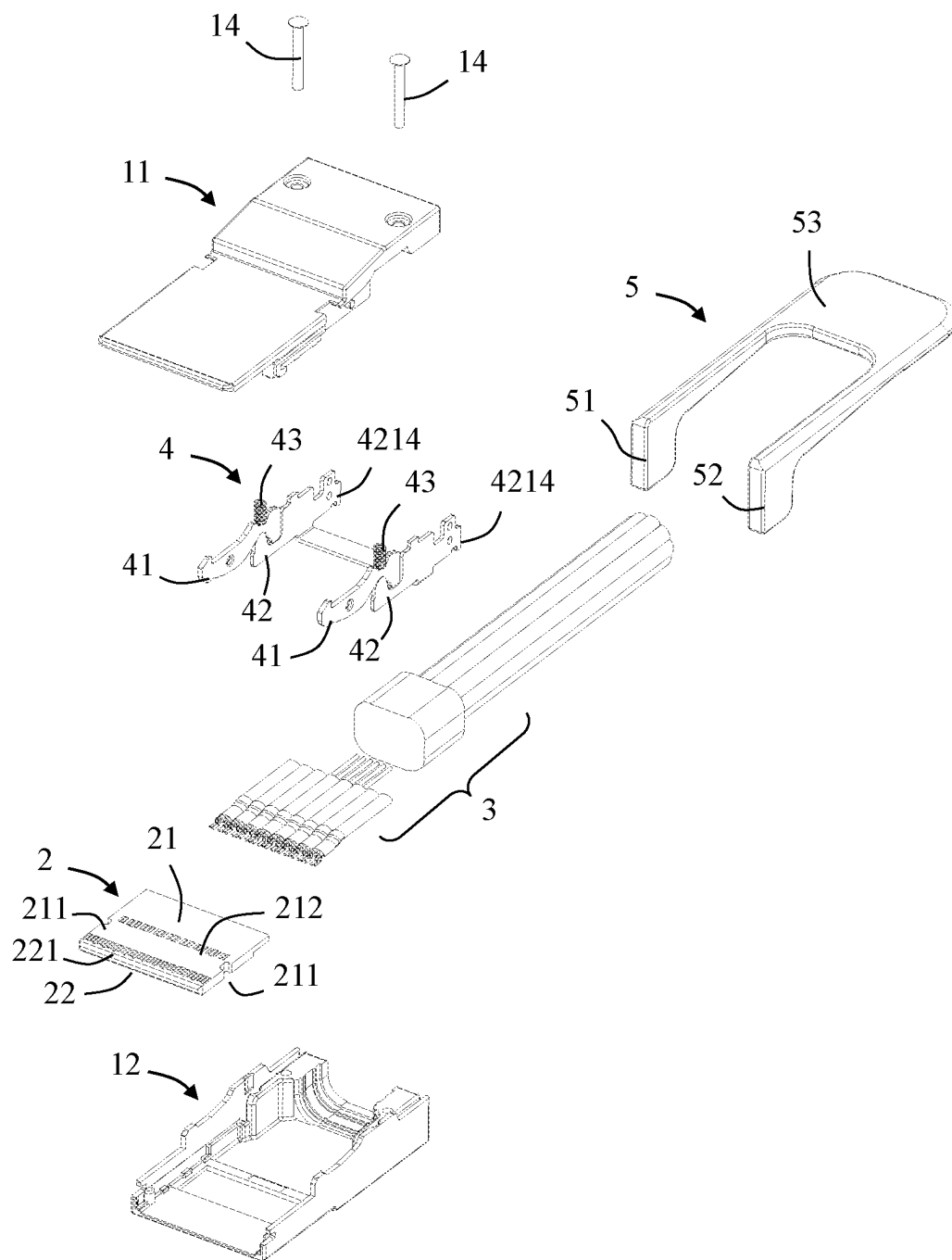


FIG. 9

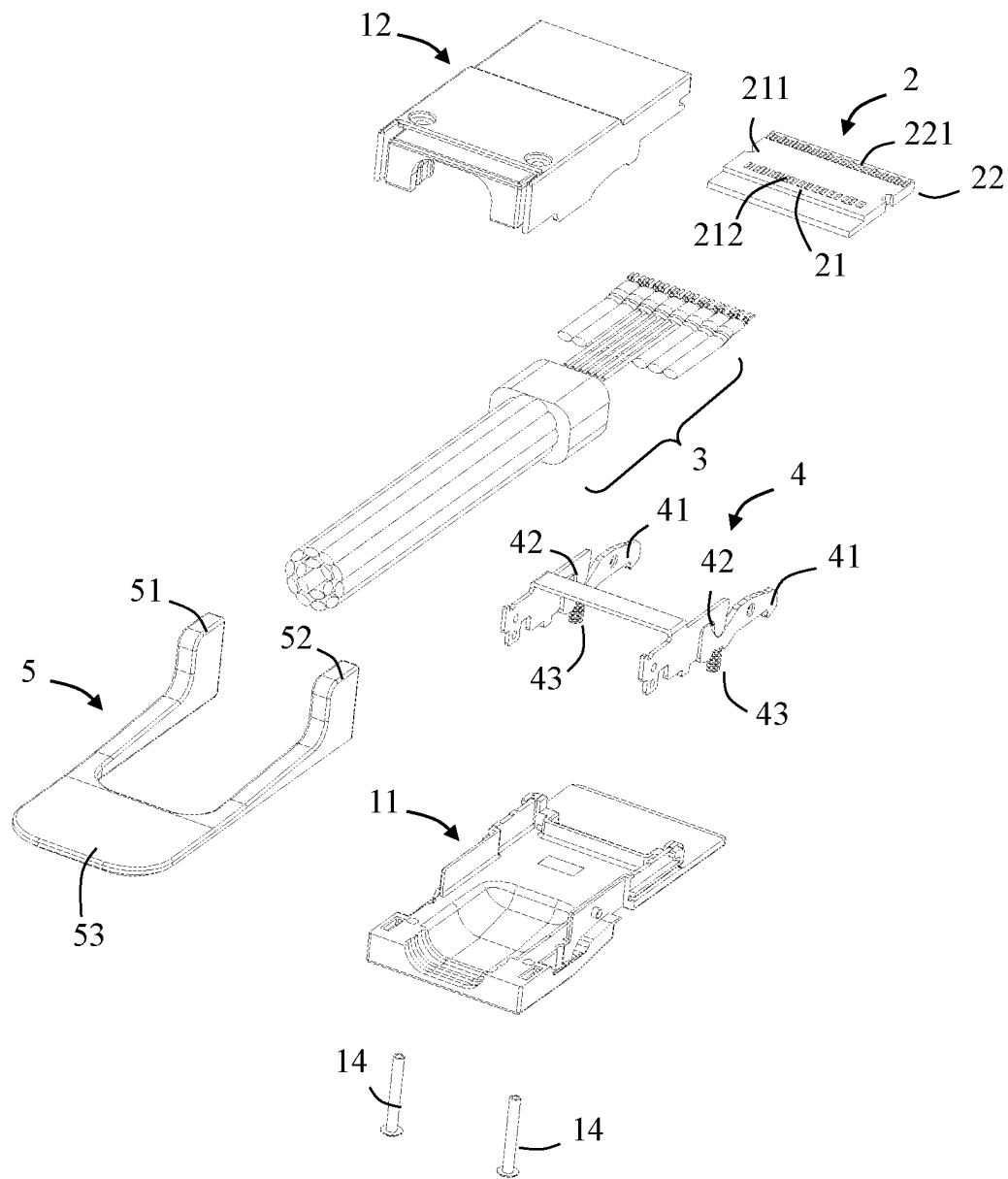


FIG. 10

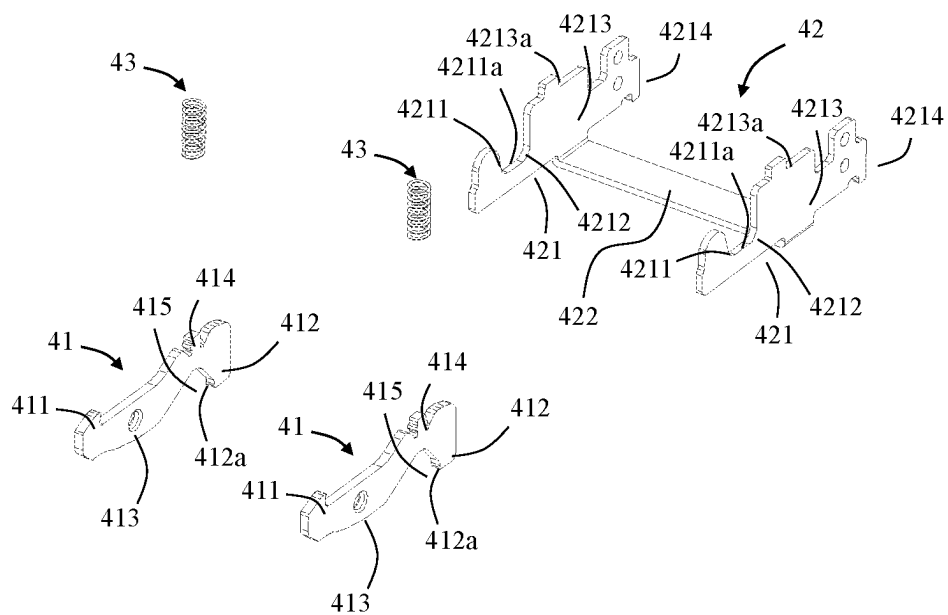


FIG. 11

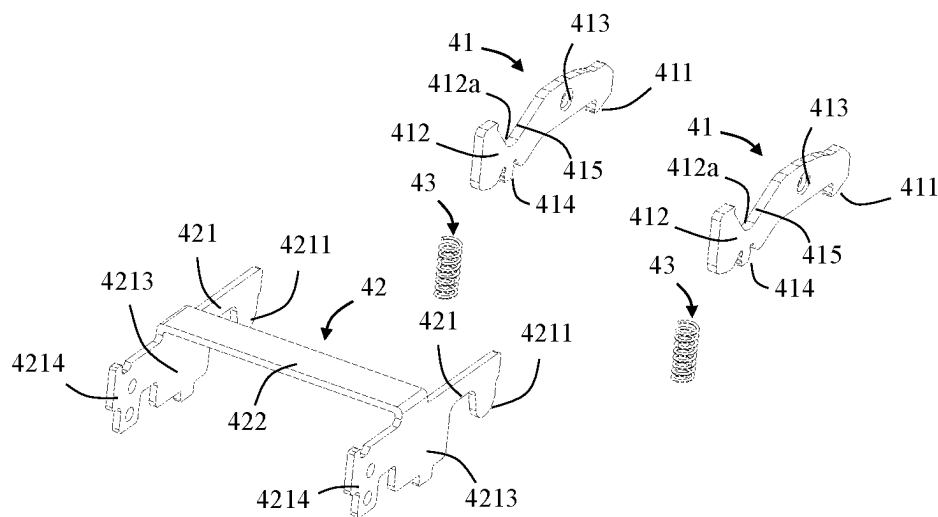


FIG. 12

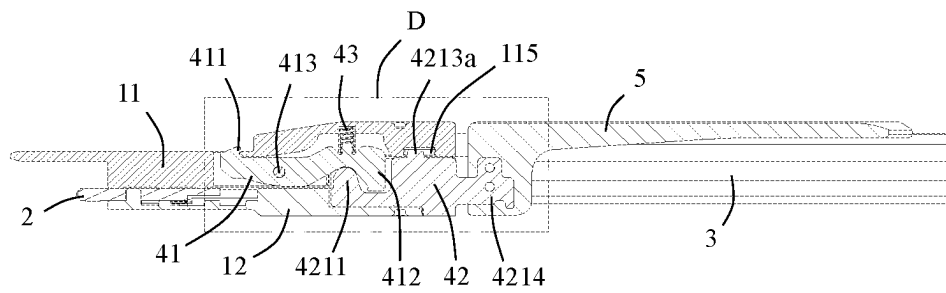


FIG. 13

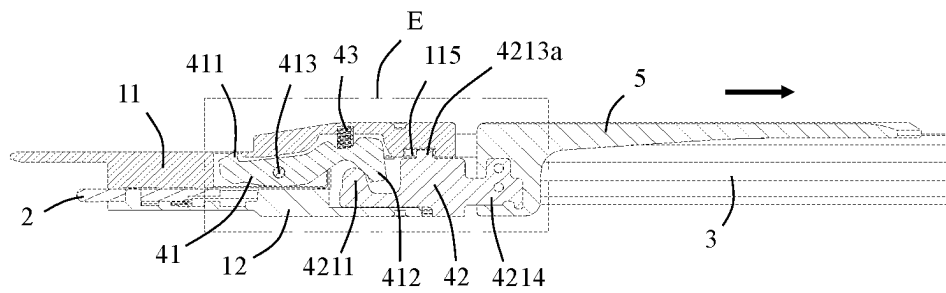


FIG. 14

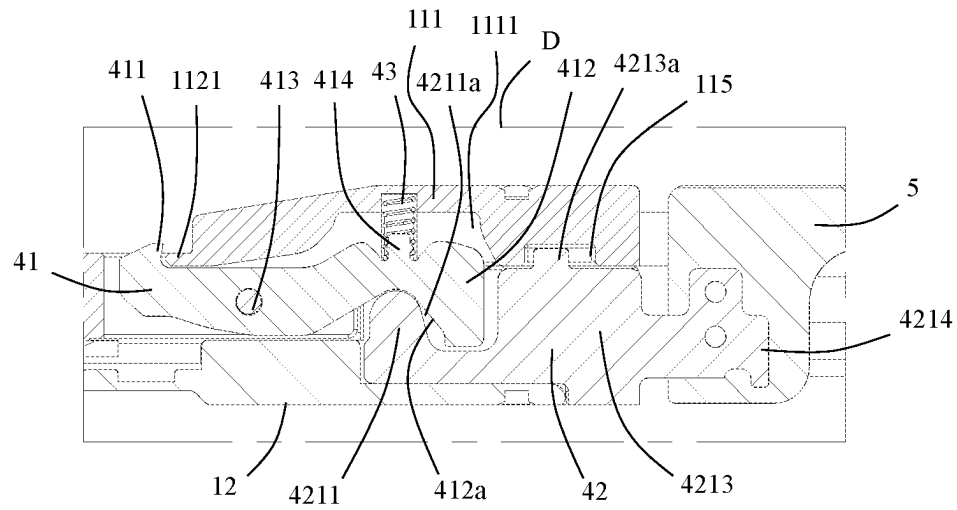


FIG. 15

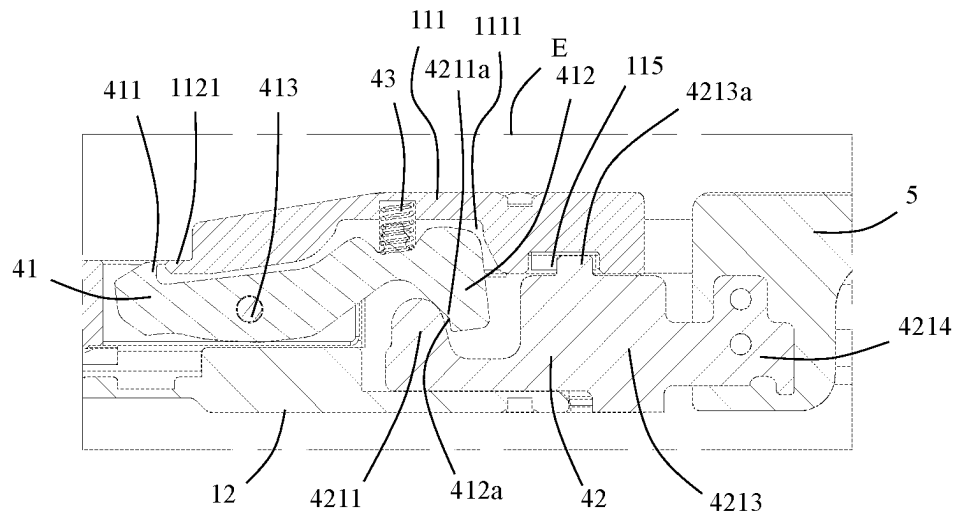


FIG. 16

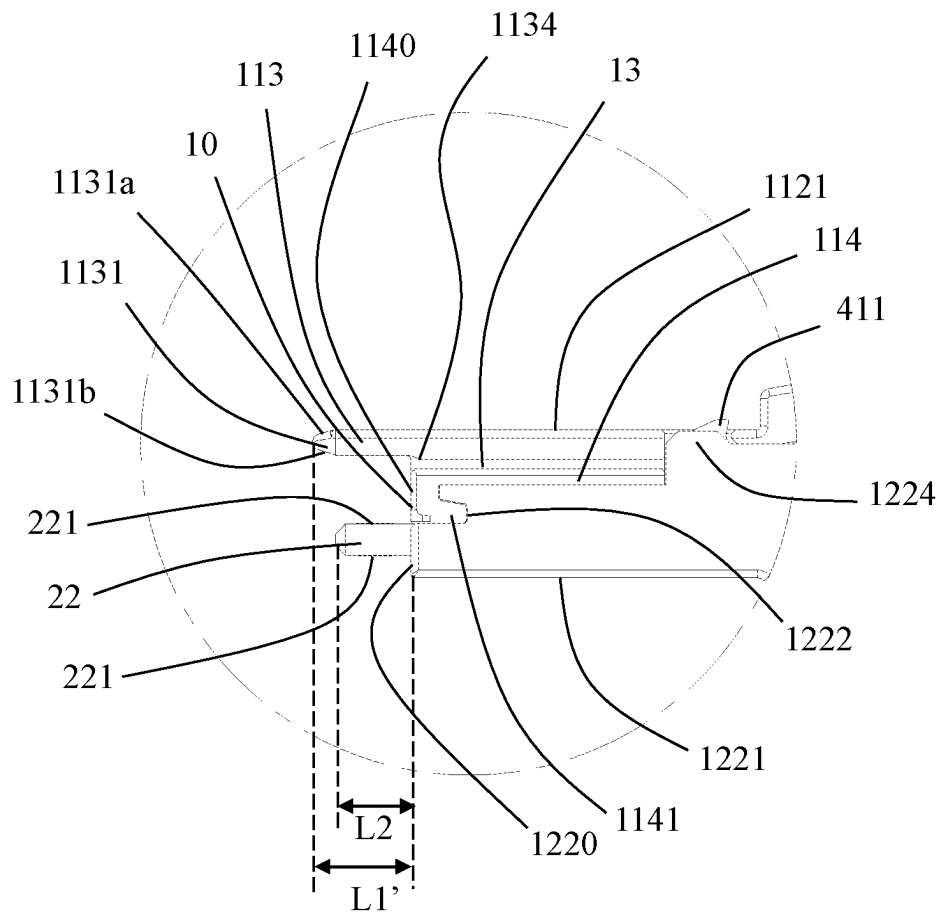


FIG. 17

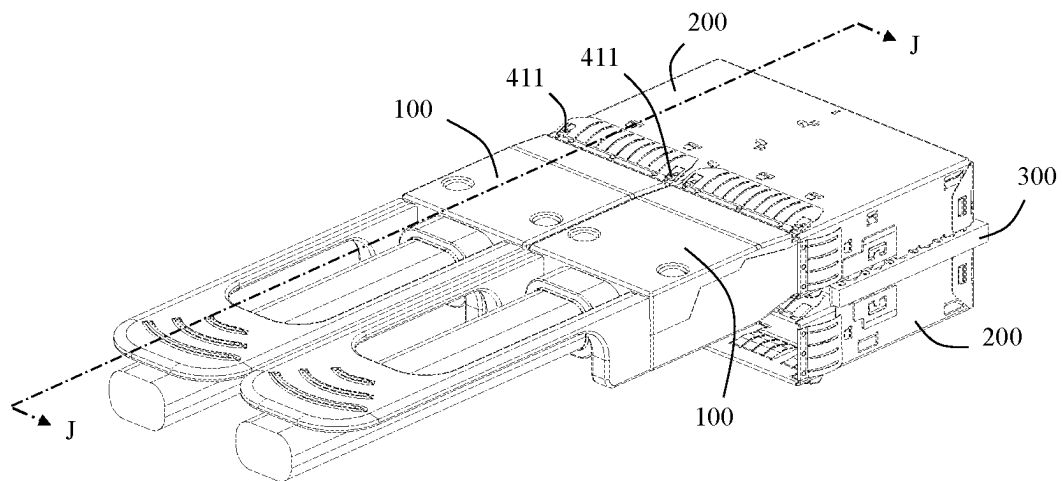


FIG. 18

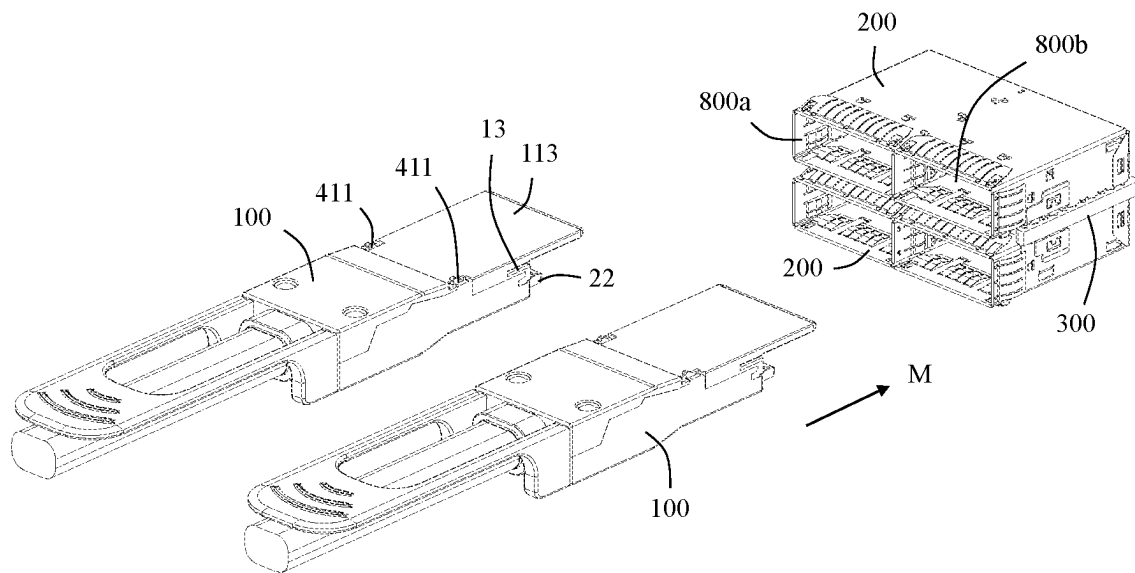


FIG. 19

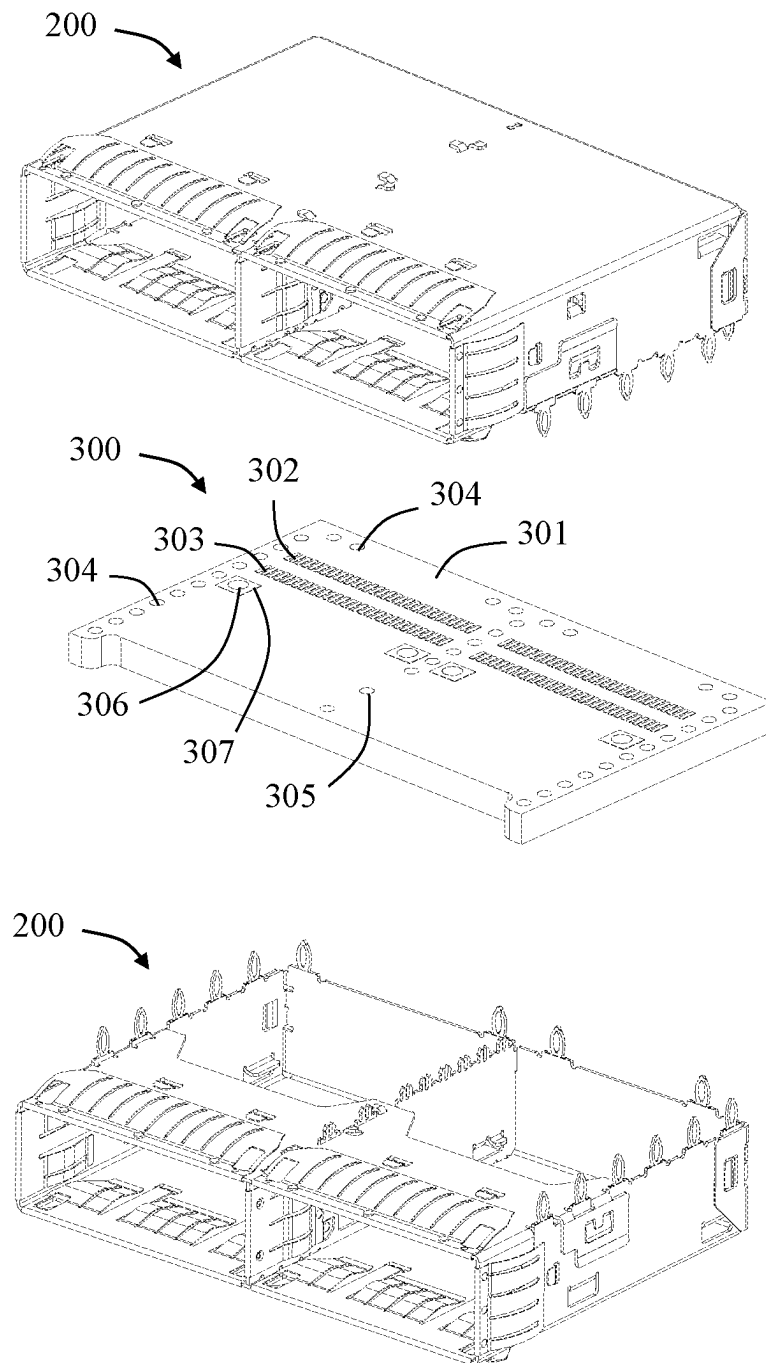


FIG. 20

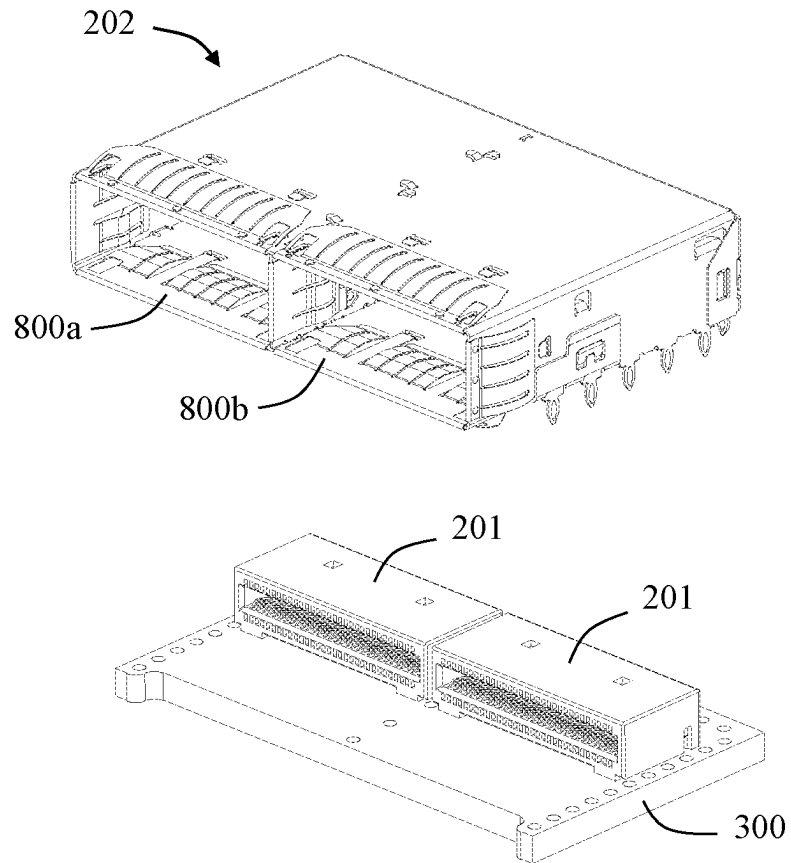


FIG. 21

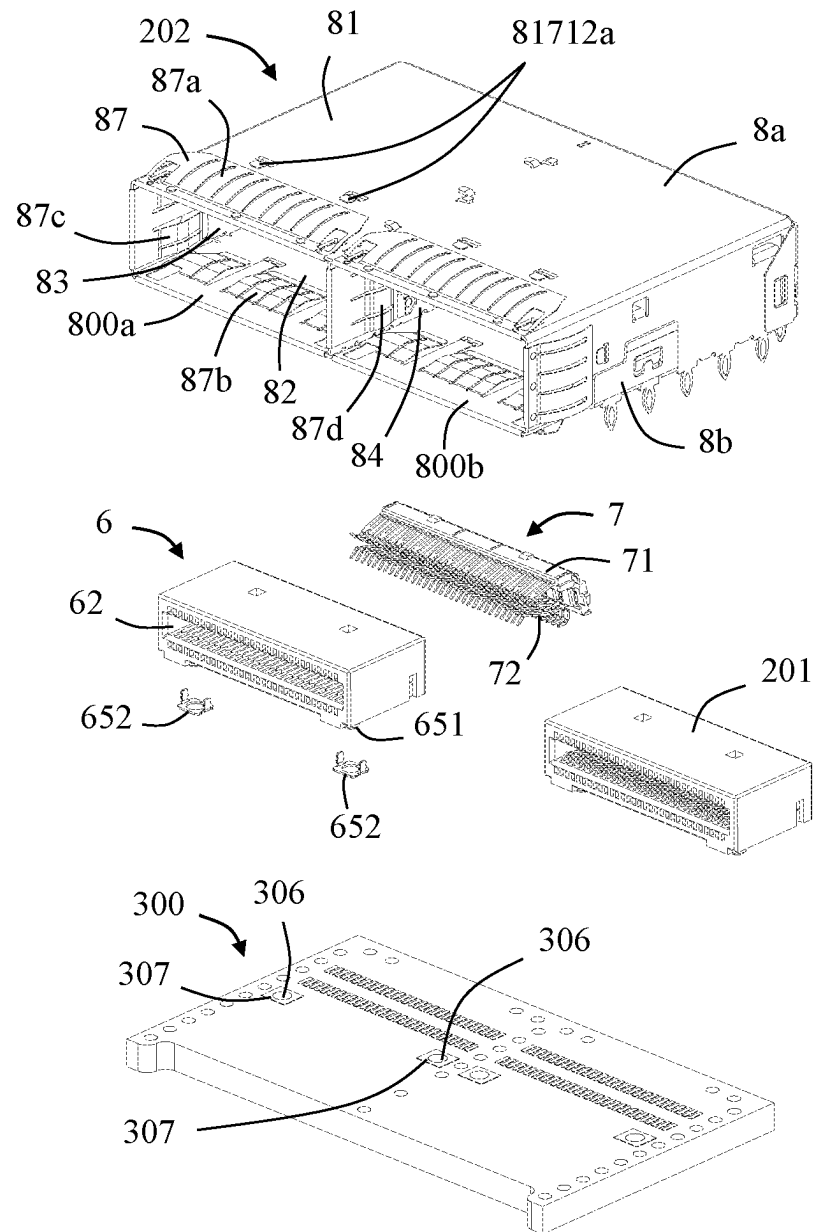


FIG. 22

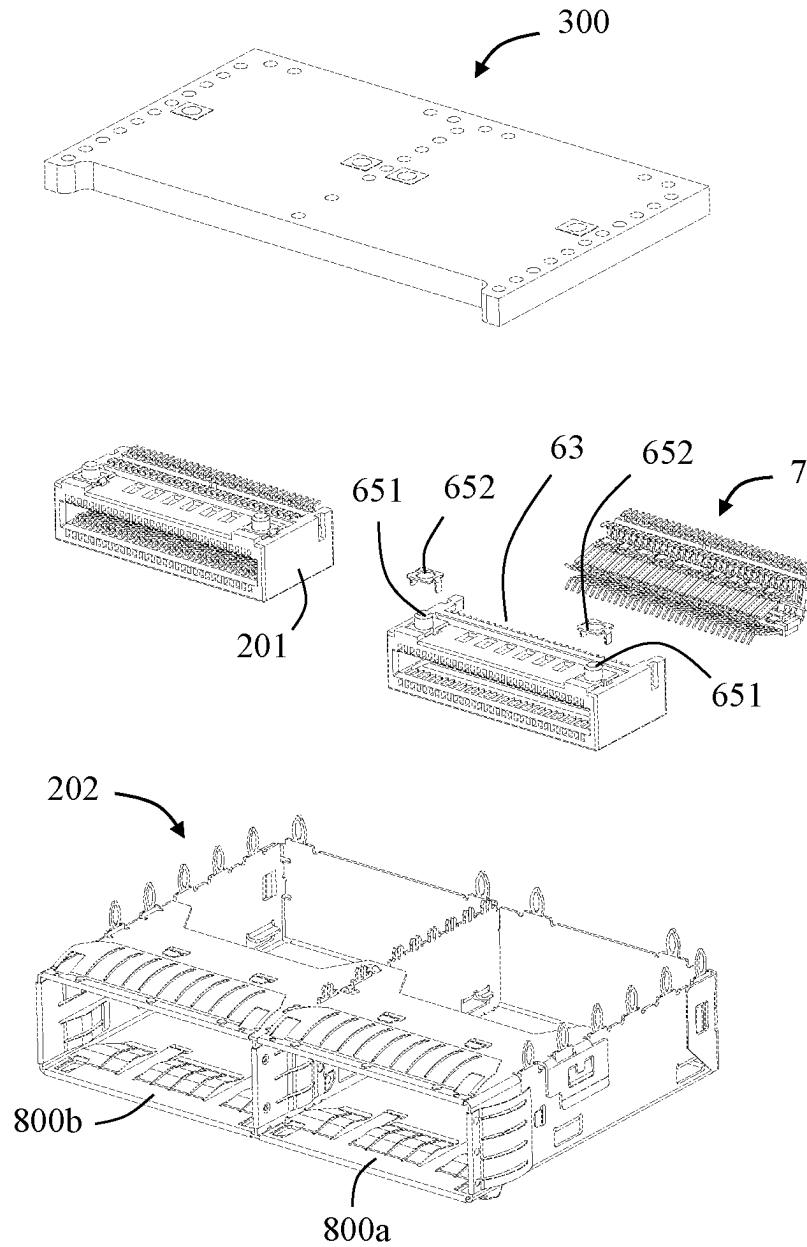


FIG. 23

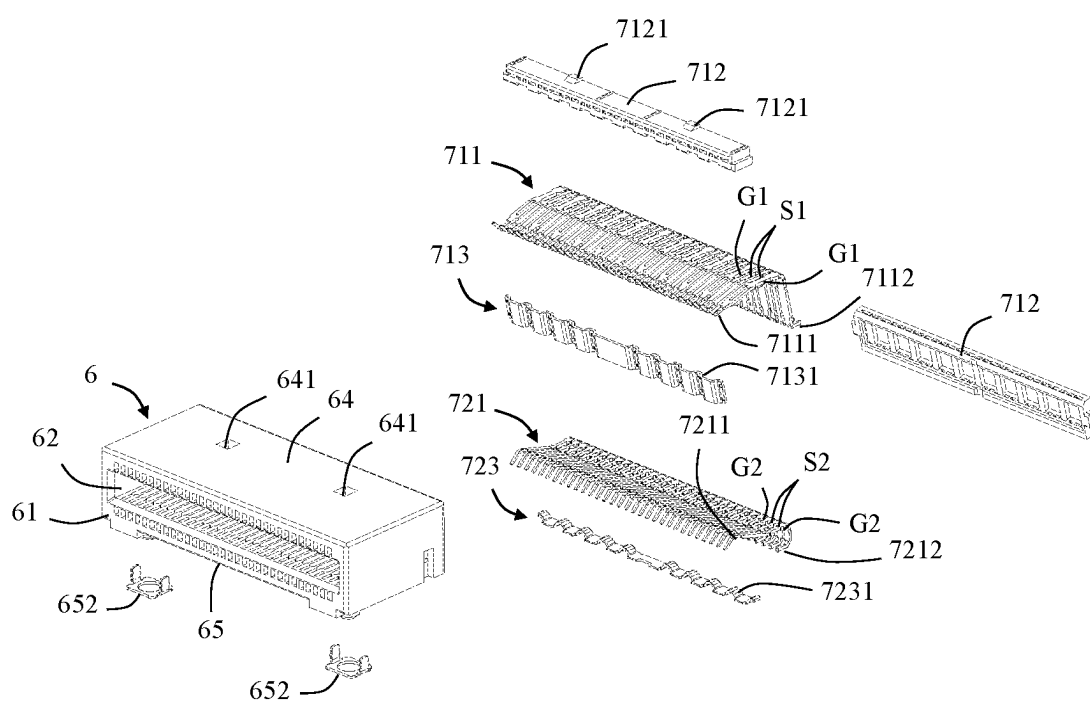


FIG. 24

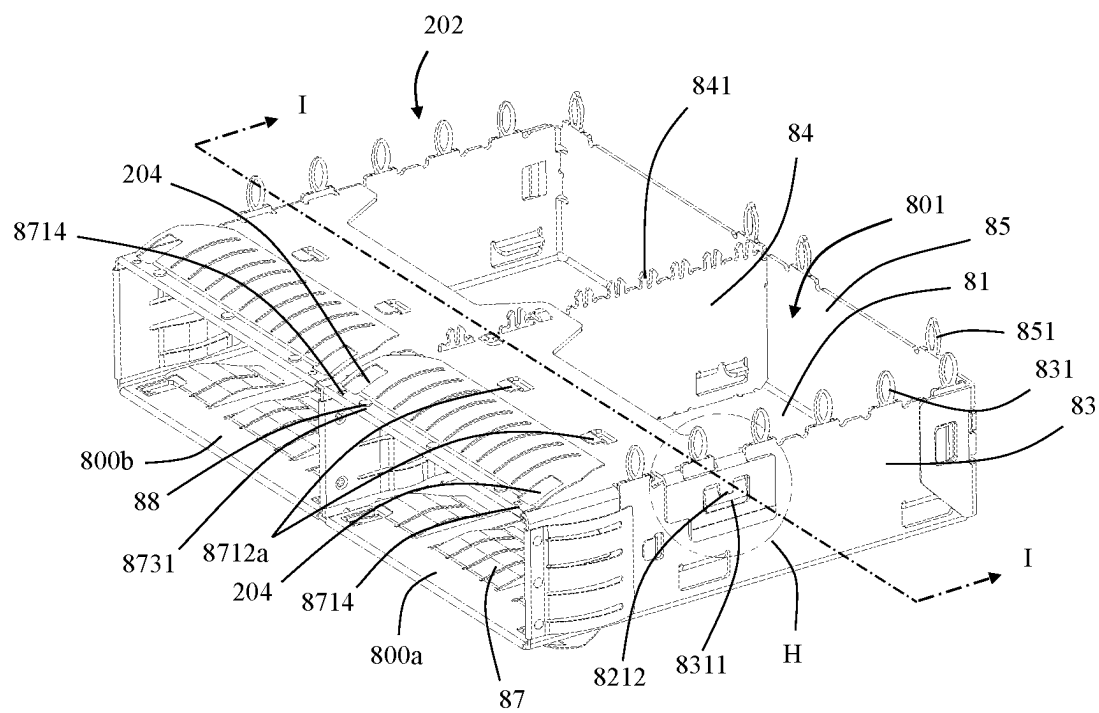


FIG. 25

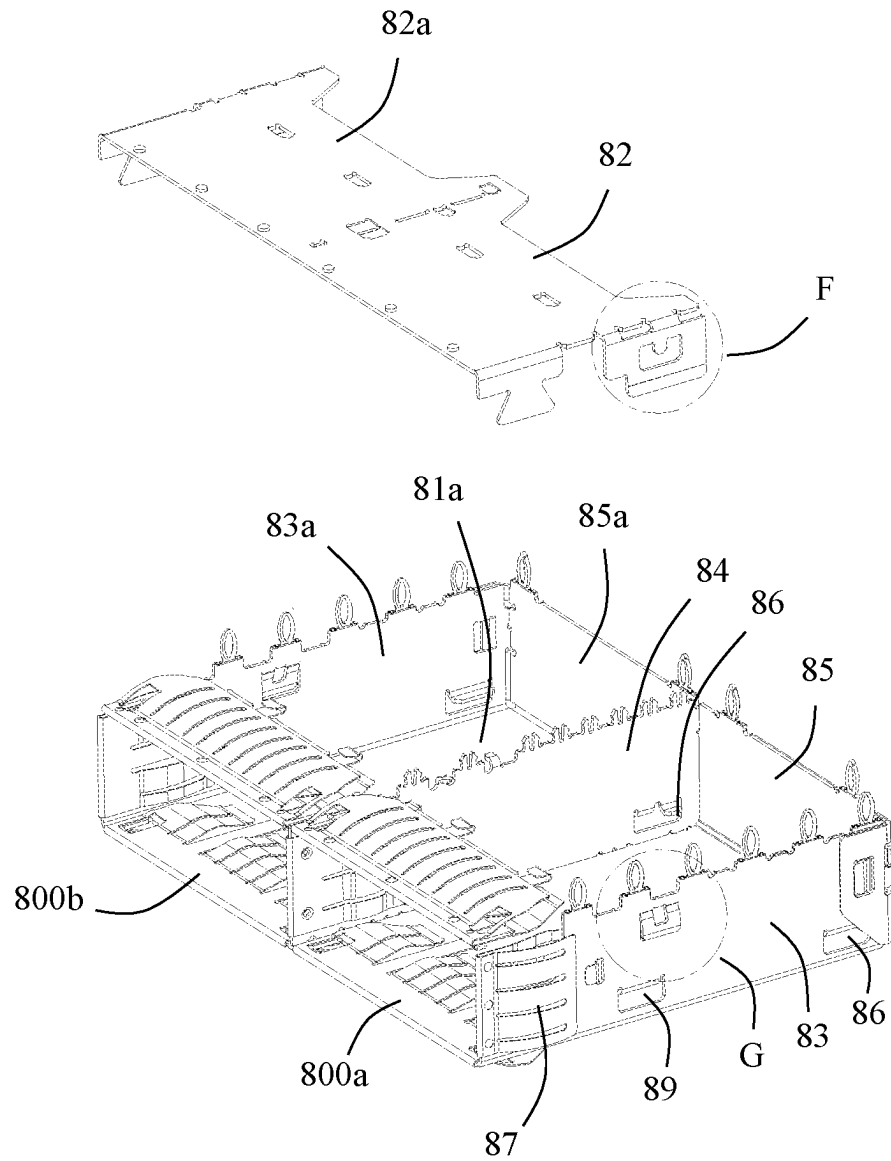


FIG. 26

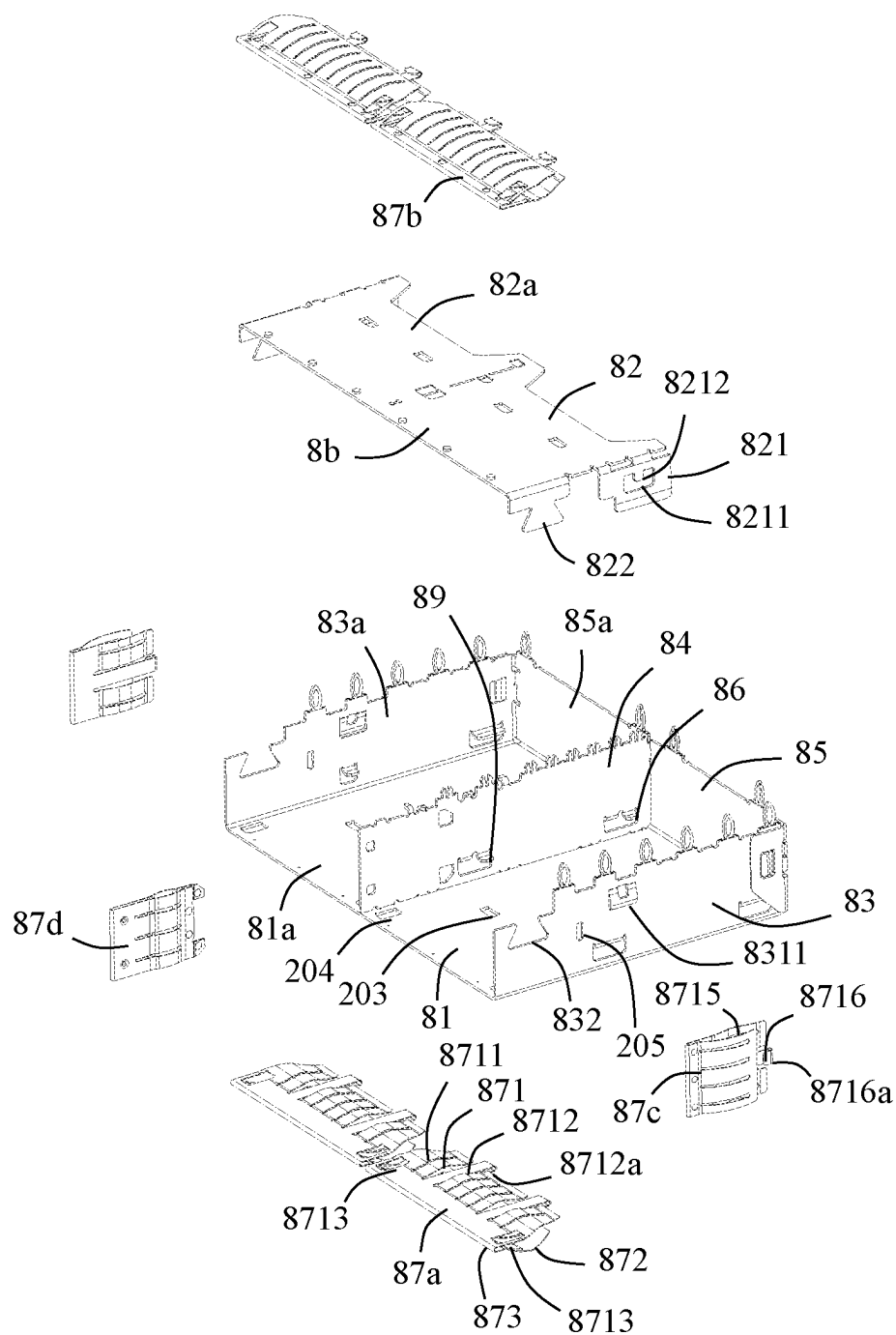


FIG. 27

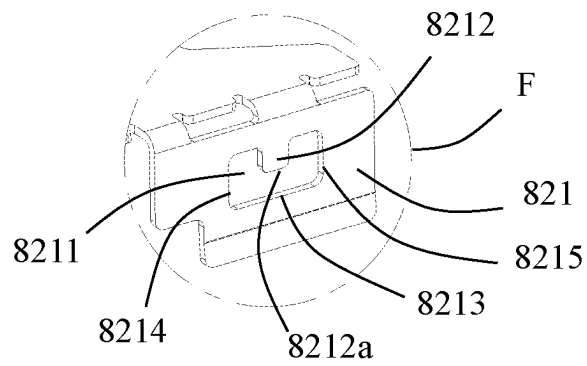


FIG. 28

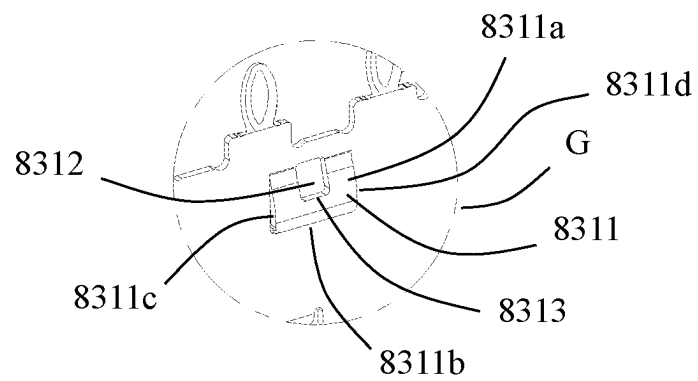


FIG. 29

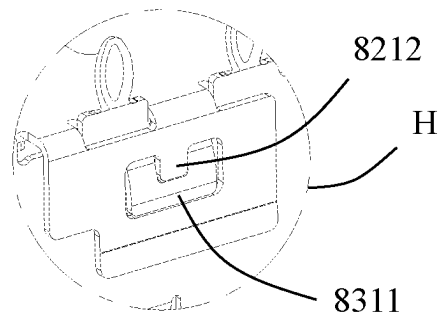


FIG. 30

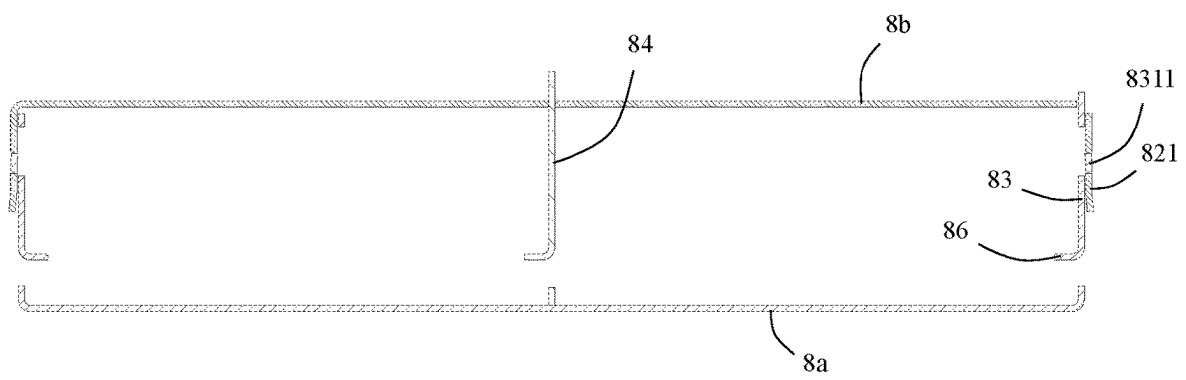


FIG. 31

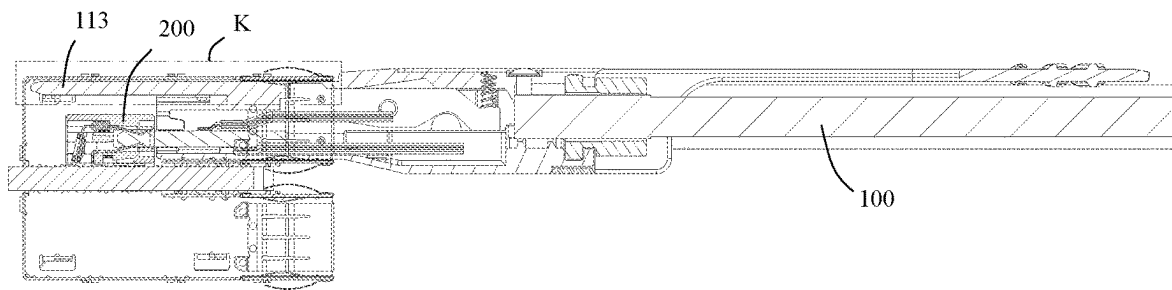


FIG. 32

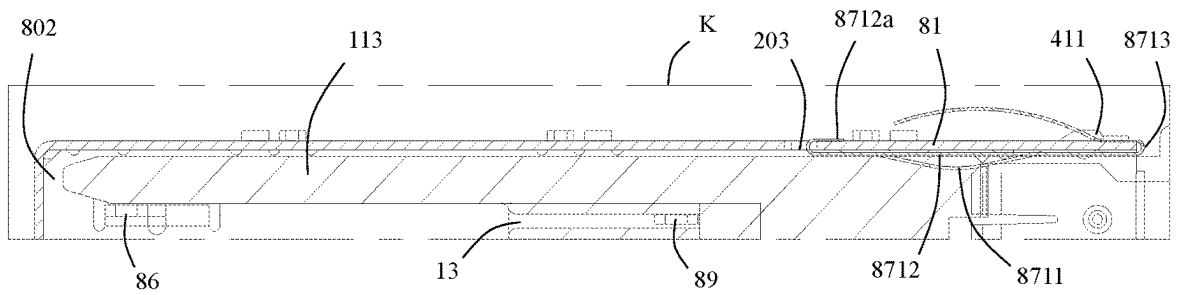


FIG. 33

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PLUG CONNECTOR AND CONNECTOR ASSEMBLY CAPABLE OF ACHIEVING MINIATURIZATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application is a continuation of U.S. patent application Ser. No. 17/979,604, filed on Nov. 2, 2022 and titled "RECEPTACLE CONNECTOR ASSEMBLY AND CONNECTOR ASSEMBLY WITH IMPROVED INSERTION RELIABILITY", which claims priority of a Chinese Patent Application No. 202210371851.1, filed on Apr. 11, 2022 and titled "RECEPTACLE CONNECTOR ASSEMBLY AND CONNECTOR ASSEMBLY", the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a plug connector and a connector assembly, which belongs to a technical field of connectors.

BACKGROUND

A connector assembly in the related art generally includes a receptacle connector and a plug connector mated with the receptacle connector. The plug connector includes a housing, a tongue plate and a cable. A plurality of conductive pads are provided on at least one surface of the tongue plate. The plug connector includes a mating surface. The tongue plate generally extends beyond the mating surface. The housing includes a plurality of wall portions surrounding the tongue plate. By providing the plurality of wall portions, the tongue plate can be better shielded and the tongue plate can be better protected. However, this design results in a larger size of the plug connector, which is not beneficial to achieve miniaturization.

SUMMARY

An object of the present disclosure is to provide a plug connector and a connector assembly which are beneficial to achieve miniaturization.

In order to achieve the above object, the present disclosure adopts the following technical solution: a connector assembly, including: a receptacle connector assembly and a plug connector; the receptacle connector assembly including: a receptacle connector, the receptacle connector including an insulating body and a plurality of conductive terminals disposed on the insulating body; the insulating body including an insertion surface and an insertion slot extending through the insertion surface; and a metal cage, the metal cage at least partially shielding the receptacle connector; the metal cage defining a first insertion space in communication with the insertion slot; the plug connector including: a housing, the housing including a mating surface; and a tongue plate on which a plurality of conductive pads are provided; wherein the housing includes an extension plate extending along a mating direction of the plug connector; the tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction; the plug connector is provided with the extension plate which is located only on one side of the tongue plate; the extension plate extends and protrudes beyond the tongue plate along the mating direction.

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In order to achieve the above object, the present disclosure adopts the following technical solution: a plug connector, including: a housing, the housing including a mating surface; a tongue plate on which a plurality of conductive pads are provided; and a plurality of cables electrically connected to the plurality of conductive pads; wherein the housing includes an extension plate extending along a mating direction of the plug connector; the tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction; the plug connector is provided with the extension plate which is located only on one side of the tongue plate; the extension plate extends and protrudes beyond the tongue plate along the mating direction.

In order to achieve the above object, the present disclosure adopts the following technical solution: a plug connector, including: a housing, the housing including a mating surface; a built-in circuit board partially located in the housing, the built-in circuit board including a tongue plate on which a plurality of conductive pads are provided; and a plurality of cables electrically connected to the built-in circuit board; wherein the housing includes an extension plate extending along a mating direction of the plug connector; the tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction; the extension plate extends and protrudes beyond the tongue plate along the mating direction; wherein a length of the extension plate protruding beyond the mating surface along the mating direction is L1, and a length of the tongue plate protruding beyond the mating surface along the mating direction is L2, where $L1 \geq 2 * L2$.

Compared with the prior art, the housing of the present disclosure includes the extension plate extending along the mating direction of the plug connector. The tongue plate and the extension plate both extend and protrude from the mating surface along the mating direction. The plug connector is provided with the extension plate which is located only on one side of the tongue plate. The extension plate extends and protrudes beyond the tongue plate along the mating direction. On the one hand, the present disclosure facilitates miniaturization of the plug connector; and on the other hand, providing the extension plate also helps to improve the stability when the plug connector is inserted into the receptacle connector assembly.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective schematic view of a plug connector in accordance with an embodiment of the present disclosure;

FIG. 2 is a perspective schematic view of FIG. 1 from another angle;

FIG. 3 is a right side view of FIG. 1;

FIG. 4 is a left side view of FIG. 1;

FIG. 5 is a partial enlarged view of a circled part A in FIG. 1;

FIG. 6 is a partial enlarged view of a circled part B in FIG. 3;

FIG. 7 is a partially exploded perspective view of FIG. 1;

FIG. 8 is a partial perspective exploded view of FIG. 7 from another angle;

FIG. 9 is a further partial perspective exploded view of FIG. 7;

FIG. 10 is a partial perspective exploded view of FIG. 9 from another angle;

FIG. 11 is a perspective exploded view of two return springs, two locking arms and a driving member in FIG. 9;

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FIG. 12 is a perspective exploded view of FIG. 11 from another angle;

FIG. 13 is a schematic cross-sectional view taken along line C-C in FIG. 1, wherein the locking arm is located at a locked position;

FIG. 14 is a schematic cross-sectional view of another state in FIG. 13, wherein the locking arm is located at an unlocked position;

FIG. 15 is a partial enlarged view of a frame part D in FIG. 13;

FIG. 16 is a partial enlarged view of a frame part E in FIG. 14;

FIG. 17 is a partial enlarged view of the circled portion in FIG. 6 in another embodiment of the present disclosure;

FIG. 18 is a schematic perspective view of a connector assembly in accordance with an embodiment of the present disclosure, wherein two plug connectors are inserted into a receptacle connector assembly;

FIG. 19 is an exploded perspective view of FIG. 18;

FIG. 20 is a perspective view of the present disclosure when the two receptacle connector assemblies are separated from a circuit board;

FIG. 21 is a schematic perspective view of the receptacle connector of the present disclosure when the receptacle connector is mounted on the circuit board, and the metal cage is separated from the receptacle connector;

FIG. 22 is a further partial perspective exploded view of FIG. 21;

FIG. 23 is a partially exploded perspective view of FIG. 22 from another angle;

FIG. 24 is a further exploded perspective view of the receptacle connector in FIG. 22;

FIG. 25 is a perspective view of the metal cage in FIG. 22 from another angle;

FIG. 26 is a partial perspective exploded view of FIG. 25;

FIG. 27 is a further partial perspective exploded view of FIG. 26;

FIG. 28 is a partial enlarged view of a circled portion F in FIG. 26;

FIG. 29 is a partial enlarged view of a circled part G in FIG. 26;

FIG. 30 is a partial enlarged view of a circled portion H in FIG. 25;

FIG. 31 is a schematic cross-sectional view taken along line I-I in FIG. 25;

FIG. 32 is a schematic cross-sectional view taken along line J-J in FIG. 18; and

FIG. 33 is a partial enlarged view of a frame portion K in FIG. 32.

DETAILED DESCRIPTION

Exemplary embodiments will be described in detail here, examples of which are shown in drawings. When referring to the drawings below, unless otherwise indicated, same numerals in different drawings represent the same or similar elements. The examples described in the following exemplary embodiments do not represent all embodiments consistent with this application. Rather, they are merely examples of devices and methods consistent with some aspects of the application as detailed in the appended claims.

The terminology used in this application is only for the purpose of describing particular embodiments, and is not intended to limit this application. The singular forms “a”, “said”, and “the” used in this application and the appended claims are also intended to include plural forms unless the context clearly indicates other meanings.

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It should be understood that the terms “first”, “second” and similar words used in the specification and claims of this application do not represent any order, quantity or importance, but are only used to distinguish different components.

Similarly, “an” or “a” and other similar words do not mean a quantity limit, but mean that there is at least one; “multiple” or “a plurality of” means two or more than two. Unless otherwise noted, “front”, “rear”, “lower” and/or “upper” and similar words are for ease of description only and are not limited to one location or one spatial orientation. Similar words such as “include” or “comprise” mean that elements or objects appear before “include” or “comprise” cover elements or objects listed after “include” or “comprise” and their equivalents, and do not exclude other elements or objects. The term “a plurality of” mentioned in the present disclosure includes two or more.

Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the case of no conflict, the following embodiments and features in the embodiments can be combined with each other.

Referring to FIGS. 1 to 10, the present disclosure discloses a plug connector 100 including a housing 1, a built-in circuit board 2 partly located in the housing 1, a cable 3 electrically connected to the built-in circuit board 2, a locking mechanism 4 installed in the housing 1, and a pull strap 5 connected with the locking mechanism 4. In an embodiment of the present disclosure, the plug connector 100 is a high-speed interconnect plug connector, including but not limited to, a plug connector based on SFP (Small Form Factor Pluggable), QSFP (Quad Small Form Factor Pluggable), OSFP (Octal Small Form Factor Pluggable), QSFP-DD (Quad Small Form Factor Pluggable-Double Density), SFP-DD (Small Form Factor Pluggable-Double Density) or DSFP (Dual Chisel Small Form-factor Pluggable). The plug connector 100 is adapted to mate with a receptacle connector assembly 200 along a mating direction M so as to realize high-speed data transmission. Correspondingly, a receptacle connector 201 of the receptacle connector assembly 200 is a receptacle connector based on SFP (Small Form Factor Pluggable), QSFP (Quad Small Form Factor Pluggable), OSFP (Octal Small Form Factor Pluggable), QSFP-DD (Quad Small Form Factor Pluggable-Double Density), SFP-DD (Small Form Factor Pluggable-Double Density) or DSFP (Dual Chisel Small Form-factor Pluggable).

Referring to FIGS. 1 to 6, in the illustrated embodiment of the present disclosure, the housing 1 includes a first housing 11 and a second housing 12. The housing 1 includes a mating surface 10 and two slots 13 located on opposite sides (for example, a left side and a right side) of the plug connector 100 and extending through the mating surface 10. The slots 13 can be adapted to mate with first guide protrusions 89 of the receptacle connector 201 (shown in FIG. 27), which is beneficial to improve the insertion reliability of the plug connector 100.

Referring to FIGS. 7 and 8, the first housing 11 is an integral structure and includes a first base 111, a first extension portion 112 extending from the first base 111, and an extension plate 113 extending from the first extension portion 112. As shown in FIG. 15 and FIG. 16, the first base 111 is located in an avoidance space 1111 at a bottom of the first base 111. The first extension portion 112 includes a first surface 1121 (for example, an upper surface) and two openings 1122 extending through the first surface 1121 along a thickness direction T-T of the plug connector 100. The two openings 1122 are located on opposite sides (for

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example, a left side and a right side) of the first extension portion 112 along a width direction W-W of the plug connector 100. Referring to FIG. 7, it is understandable to those skilled in the art that in the illustrated embodiment of the present disclosure, the mating direction M is a rear-to-front direction, the width direction W-W is a left-right direction, and the thickness direction is a vertical direction. The mating direction M, the width direction W-W and the thickness direction T-T are perpendicular to each other. In the illustrated embodiment of the present disclosure, the two openings 1122 are spaced apart along the width direction W-W. The two openings 1122 extend through two sides of the first extension portion 112 of the housing 1, respectively. As shown in FIG. 6, in the illustrated embodiment of the present disclosure, the extension plate 113 includes a guide portion 1131 located at an end of the extension plate 113 along the mating direction M. An upper surface 1131a of the guide portion 1131 and/or a lower surface 1131b of the guide portion 1131 is a guide inclined surface. In the illustrated embodiment of the present disclosure, the upper surface 1131a of the guide portion 1131 and the lower surface 1131b of the guide portion 1131 are both guide inclined surfaces. Besides, the extension plate 113 includes an upper surface 1132 and a lower surface 1133 opposite to the upper surface 1132 of the extension plate 113. In the illustrated embodiment of the present disclosure, the first extension portion 112 includes two convex portions 114 which protrude downwardly along the thickness direction T-T and are located on opposite sides (for example, a left side and a right side) of the first extension portion 112. Each convex portion 114 extends along the mating direction M. The convex portion 114 includes a first front end surface 1140 and a locking portion 1141 located at a front portion of the convex portion 114. In the illustrated embodiment of the present disclosure, the first front end surface 1140 is a vertical surface. The locking portion 1141 is of a U-shaped configuration.

The second housing 12 includes a second base 121 and a second extension portion 122 extending from the second base 121. The first base 111 corresponds to the second base 121, wherein the first base 111 is located above the second base 121 as a whole, and the first base 111 and the second base 121 are aligned in the vertical direction. The first extension portion 112 corresponds to the second extension portion 122, wherein the first extension portion 112 is located above the second extension portion 122 as a whole, and the first extension portion 112 and the second extension portion 122 are aligned in the vertical direction. The second extension portion 122 includes a second front end surface 1220. The first front end surface 1140 and the second front end surface 1220 are aligned with each other in the thickness direction T-T to jointly form the mating surface 10. The second extension portion 122 includes a second surface 1221 (for example, a lower surface). The first surface 1121 and the second surface 1221 are disposed opposite to each other along the thickness direction T-T. Referring to FIG. 6, in the illustrated embodiment of the present disclosure, the upper surface 1132 of the extension plate 113 is coplanar with the first surface 1121 of the first extension portion 112. A front side of the second extension portion 122 includes a locking slot 1222. In the illustrated embodiment of the present disclosure, the locking slot 1222 is of a U-shaped configuration. The locking portion 1141 is fixed in the locking slot 1222 to prevent the first housing 11 and the second housing 12 from being separated from each other in the thickness direction T-T. The second extension portion 122 further includes two positioning posts 1223 on opposite sides. The positioning posts 1223 are adapted for positioning

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the built-in circuit board 2. A rear side of the second extension portion 122 is further provided with a limiting protrusion 1224 which protrudes into a corresponding opening 1122 along the thickness direction T-T.

In the illustrated embodiment of the present disclosure, the plug connector 100 further includes a bolt 14 to assemble and fix the first base 111 and the second base 121. Of course, in other embodiments, the first housing 11 and the second housing 12 may be assembled through a mutual locking structure (for example, a locking arm and a locking groove for mating with the locking arm).

Referring to FIGS. 7, 9 and 10, the built-in circuit board 2 includes a base plate 21 at least partially clamped between the first extension portion 112 and the second extension portion 122, and a tongue plate 22 extending forward from the base plate 21. Opposite sides of the base plate 21 include two positioning notches 211 matched with the positioning posts 1223. At least one surface of the base plate 21 is provided with a plurality of soldering pads 212 which are adapted for being soldered and fixed with the cable 3. In the illustrated embodiment of the present disclosure, two opposite surfaces (for example, an upper surface and a lower surface) of the base plate 21 are respectively provided with the soldering pads 212, which is beneficial to maximize the use of the space of the base plate 21 as much as possible. A plurality of conductive pads 221 are provided on at least one surface of the tongue plate 22. The conductive pads 221 are adapted to contact the conductive terminals of the receptacle connector 201 so as to realize data transmission. In the illustrated embodiment of the present disclosure, the two opposite surfaces (for example, the upper surface 223 and the lower surface 224) of the tongue plate 22 are respectively provided with the conductive pads 221, which is beneficial to maximize the use of the space of the tongue plate 22 as much as possible.

As shown in FIGS. 3 and 6, both the tongue plate 22 and the extension plate 113 protrude beyond the mating surface 10 along the mating direction M. The extension plate 113 further protrudes beyond the tongue plate 22 along the mating direction M. In the illustrated embodiment of the present disclosure, only one extension plate 113 is provided on one side of the tongue plate 22 along the thickness direction T-T of the plug connector 100. In other words, an extension plate 113 is provided on one side of the tongue plate 22 along the thickness direction T-T of the plug connector 100. However, on an opposite side of the tongue plate 22 opposite to the extension plate 113, there is no structure protruding beyond the mating surface 10 along the mating direction M. In the illustrated embodiment of the present disclosure, a length of the extension plate 113 protruding beyond the mating surface 10 along the mating direction M is L1, a length of the tongue plate 22 protruding beyond the mating surface along the mating direction M is L2, where $L1 \geq 2 * L2$. With this arrangement, by appropriately increasing the length of the extension plate 113, when the plug connector 100 is inserted into the receptacle connector assembly 200, the extension plate 113 will contact the receptacle connector assembly 200 as early as possible to achieve pre-positioning. This is also beneficial to protect the tongue plate 22 and avoid possible damage to the tongue plate 22 due to incorrect insertion angles or mismatched receptacle connectors. In addition, this design is also beneficial to increase the insertion depth of the plug connector 100 and improve the mating reliability of the plug connector 100 with the receptacle connector 201 when the plug connector 100 is inserted in place. Referring to FIG. 17, in another embodiment of the plug connector 100 of the

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present disclosure, a length of the extension plate 113 protruding beyond the mating surface 10 along the mating direction M is L1', a length of the tongue plate 22 protruding beyond the mating surface along the mating direction M is L2, where L1'>L2. Compared to the embodiment in FIG. 6, the length L1' of the plug connector 100 in FIG. 17 is slightly greater than the length L2. Even so, by protruding the extension plate 113 out of the tongue plate 22 in the mating direction M, the extension plate 113 can still provide protection to the tongue plate 22 when the plug connector 100 is inserted into the receptacle connector assembly 200. It prevents the tongue plate 22 from firstly touching components of the receptacle connector assembly 200, thereby reducing the risk of damage to the tongue plate 22. In an embodiment of the present disclosure, a value range of L1 is 12.0 mm to 13.0 mm, and a value range of L2 is 3.8 mm to 3.9 mm. In an embodiment of the present disclosure, a width of the extension plate 113 along the width direction W-W of the plug connector 100 ranges from 20.0 mm to 21.0 mm. A width of the tongue plate 22 along the width direction W-W ranges from 18.0 mm to 19.0 mm. The width of the extension plate 113 is slightly larger than the width of the tongue plate 22, so that the extension plate 113 can protect the tongue plate 22 to a certain extent.

Referring to FIG. 6, in the illustrated embodiment of the present disclosure, a distance between the upper surface 223 of the tongue plate 22 and the lower surface 1133 of the extension plate 113 along the thickness direction T-T is greater than a distance between the lower surface 224 of the tongue plate 22 and the second surface 1221 of the first extension portion 112 along the thickness direction T-T. In other words, the tongue plate 22 is located at a middle and lower portion of the housing 1 along the thickness direction T-T. In the illustrated embodiment of the present disclosure, the plug connector 100 is provided with the extension plate 113 only on one side of the tongue plate 22, which is beneficial to reduce the dimensions (such as height and width) of the plug connector 100 as much as possible, so that it is beneficial to achieve miniaturization.

The slot 13 is provided on the first extension portion 112; or the slot 13 is provided on the second extension portion 122; or the slot 13 is formed between the first extension portion 112 and the second extension portion 122 in the thickness direction T-T of the plug connector 100. In the illustrated embodiment of the present disclosure, the slot 13 is provided on the convex portion 114. The slot 13 extends backwardly to communicate with the corresponding opening 1122. A distance between the slot 13 and the first surface 1121 in the thickness direction T-T is smaller than a distance between the slot 13 and the second surface 1221. In other words, the slot 13 is arranged on an upper side.

Referring to FIG. 6, in the illustrated embodiment of the present disclosure, the housing 1 includes a slot top surface 131 downwardly exposed in the slot 13. The slot top surface 131 is lower than the lower surface 1133 of the extension plate 113 along the thickness direction T-T. The housing 1 further includes a transition inclined surface 1134 connecting the lower surface 1133 of the extension plate 113 and the slot top surface 131.

Referring to FIGS. 9 to 16, the locking mechanism 4 includes two locking arms 41 installed on two inner and opposite sides of the second housing 12, a driving member 42 mated with the locking arms 41, and two return springs 43 mated with the locking arms 41, respectively.

Each locking arm 41 includes a locking latch 411 located at a front end of the locking arm 41, an abutting portion 412 located at a rear end of the locking arm 41, and a pivot

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portion 413 located between the locking latch 411 and the abutting portion 412. The locking arm 41 is rotatable around the pivot portion 413. Referring to FIG. 13 to FIG. 16, the pivot portion 413 is located between the extension plate 113 and the tongue plate 22 along the thickness direction T-T of the plug connector 100 so as to reduce the height of the plug connector 100 as much as possible. The slot 13 extends backwardly to be adjacent to the locking latch 411. This arrangement increases a length of each slot 13 along the mating direction M and is beneficial to increase an insertion depth of the plug connector 100, thereby improving the insertion reliability of the plug connector 100.

In the illustrated embodiment of the present disclosure, the return spring 43 is a compression spring. One end of the compression spring abuts against the top of the locking arm 41; and another end of the compression spring abuts against a lower surface of the first extension portion 112. A contact position of the locking latch 411 and the locking arm 41, and a contact position of the return spring 43 and the locking arm 41 are located on two sides of the pivot portion 413, respectively. With this arrangement, the locking arm 41 is equivalent to a seesaw of which a fulcrum is the pivot portion 413. When no external force is applied, under the action of the compression spring, the locking latch 411 is located in the corresponding opening 1122 and protrudes upwardly beyond the first surface 1121. Under this condition, the locking latch 411 can be matched with a corresponding locking structure (e.g., a locking slot 204, as shown in FIG. 27) of the receptacle connector 201, that is, this state is a locking state when the plug connector 100 is inserted into the receptacle connector.

As shown in FIG. 5, each limiting protrusion 1224 is located outside the corresponding locking latch 411 along the width direction W-W of the plug connector 100 so as to limit the locking latch 411.

In the illustrated embodiment of the present disclosure, in order to better fix the compression spring, each locking arm 41 further includes a mounting post 414. One end of the compression spring is sleeved on the mounting post 414 to prevent the compression spring from leaving its original position after being stressed.

In the illustrated embodiment of the present disclosure, the pivot portion 413 includes a pivot hole. The first housing 11 includes a pivot shaft 116 that matches with the pivot hole. The pivot shaft 116 is integrally formed on the first housing 11. The locking arm 41 is rotatable around the pivot shaft 116 under the action of the external force. Of course, in other embodiments, the pivot shaft 116 may also be integrally formed on the second housing 12; or the pivot shaft 116 is a separate shaft assembled to the first housing 11 and/or the second housing 12.

Of course, in other embodiments, the return spring 43 may also be a tension spring. One end of the tension spring is fixed to the locking arm 41, and the other end of the tension spring is fixed to the housing 1.

In addition, each locking arm 41 further includes a first recess 415 located between the abutting portion 412 and the pivot portion 413, and an inner abutting surface 412a exposed in the first recess 415. In the illustrated embodiment of the present disclosure, the first recess 415 is located below the mounting post 414 and is aligned with the mounting post 414 in the vertical direction. The inner abutting surface 412a is provided on the abutting portion 412. In the illustrated embodiment of the present disclosure, the inner abutting surface 412a is an inclined surface.

The driving member 42 is connected to the pull strap 5. The driving member 42 includes two driving arms 421

located on opposite sides and a connecting rod **422** which connects the two driving arms **421**. Each driving arm **421** is located at a rear end of the corresponding locking arm **41** and abuts against the locking arm **41**. Each driving arm **421** includes a protruding portion **4211** received in the first recess **415**, a second recess **4212** for receiving the abutting portion **412**, a base portion **4213** connected to the protruding portion **4211** and a fixing portion **4214** protruding backwardly from the base portion **4213**. The protruding portion **4211** includes an outer abutting surface **4211a** which abuts against the inner abutting surface **412a**. The abutting portion **412** and the protruding portion **4211** are at least partially overlapped in a wall thickness direction of the locking arm **41**. In the illustrated embodiment of the present disclosure, the outer abutting surface **4211a** is a curved surface. Through the cooperation of the curved surface and the inclined surface, the smoothness of the interaction between the driving arm **421** and the locking arm **41** is improved. As shown in FIGS. **15** and **16**, the base portion **4213** is provided with a protrusion **4213a** extending upwardly. The first housing **11** includes a limiting slot **115**. The protrusion **4213a** is movable in the limiting slot **115** under the action of external force. In the illustrated embodiment of the present disclosure, the fixing portion **4214** is insert-molded in the pull strap **5**. The connecting rod **422** connects the base portions **4213** of the two driving arms **421** to strengthen the structure of the driving member **42**, improve the stability of the driving member **42** when moving, and improve the synchronization of the movement of the two locking arms **41**.

As shown in FIG. **9**, the pull strap **5** includes a first connecting portion **51** and a second connecting portion **52** which are respectively connected to the fixing portions **4214** of the two locking arms **41**. The first connecting portion **51** and the second connecting portion **52** respectively extend in a vertical direction and are parallel to each other. In the illustrated embodiment of the present disclosure, the first connecting portion **51** and the second connecting portion **52** are both L-shaped. The pull strap **5** further includes a bridge portion **53** connecting the first connecting portion **51** and the second connecting portion **52**. The bridge portion **53** extends in a horizontal direction.

The pull strap **5** is movable in a direction opposite to the mating direction **M** under the pulling of the external force, so as to directly or indirectly drive the locking latches **411** to move away from the first surface **1121** in the thickness direction **T-T**. Specifically, referring to FIGS. **14** and **16**, in the illustrated embodiment of the present disclosure, when the pull strap **5** moves in the direction opposite to the mating direction **M** under the pulling of the external force, each outer abutting surface **4211a** of the protruding portion **4211** abuts against the inner abutting surface **412a** of the abutting portion **412**, and slides on the inner abutting surface **412a**. Under this condition, an upward force is generated to move the abutting portions **412** upwardly and compress the compression springs. Referring to FIG. **15** and FIG. **16**, the avoidance space **1111** at the bottom of the first base **111** provides a space for the contact portion **412** to tilt upwardly. At the same time, the locking latches **411** move downwardly to move away from the first surface **1121**. This state is an unlocked state of the plug connector **100**.

When the external force is removed, the return springs **43** release elastic force, so that the abutting portions **412** move downwardly. At the same time, the locking latches **411** move upwardly, and the locking arms **41** return to their original positions.

In the present disclosure, the two return springs **43** are provided to make the force of the locking arms **41** more even, which improves the stability of the movement of the locking arms **41**.

Referring to FIG. **7** and FIG. **8**, the first housing **11** is provided with first outer walls **11a** located on two sides, respectively. The second housing **12** is provided with first inner walls **12a** located on two sides, respectively. The first outer side walls **11a** correspond to the first inner side walls **12a**. The housing **1** includes a slot **13a** located between the first outer side wall **11a** and a corresponding first inner side wall **12a**. The locking arm **41** is received in the slot **13a**. Each latch arm **41** is plate-shaped and located in a vertical plane. Each driving arm **421** is plate-shaped and located in a vertical plane.

Referring to FIGS. **18** to **26**, the present disclosure also discloses a connector assembly which includes a circuit board **300**, a plurality of receptacle connector assemblies **200** mounted on the circuit board **300**, and the plug connectors **100** for mating with the receptacle connector assemblies **200**.

In the illustrated embodiment of the present disclosure, two receptacle connector assemblies **200** are provided and installed on opposite sides (e.g., upper and lower sides) of the circuit board **300** in a belly-to-belly manner. This arrangement is beneficial to more effectively utilize the space of the circuit board **300** and improve the installation efficiency.

Each receptacle connector assembly **200** includes a plurality of receptacle connectors **201** and a metal cage **202** which shields the receptacle connectors **201**. In an embodiment shown in the present disclosure, two receptacle connectors **201** with the same structure are provided. The following only takes one of the receptacle connectors **201** as an example for detailed description.

Referring to FIG. **20**, corresponding to one receptacle connector **201**, the circuit board **300** includes a first surface **301** (e.g., an upper surface), a plurality of first soldering pads **302** on the first surface **301**, a plurality of second soldering pads **303** on the first surface **301**, a plurality of conductive holes **304** extending through the first surface **301**, a plurality of mounting holes **305** extending through the first surface **301**, a plurality of positioning through holes **306** extending through the first surface **301**, and a plurality of ground soldering pads **307** surrounding the positioning through holes **306**. In the embodiment shown in the present disclosure, the plurality of first soldering pads **302** are arranged in a first row, the plurality of second soldering pads **303** are arranged in a second row, and the first row is parallel to the second row. The positioning through holes **306** are located at the front end of the second soldering pads **303**. The conductive holes **304** are located on the left and rear sides of the first soldering pads **302** and the second soldering pads **303**. The mounting holes **305** are located on the right side of the first soldering pads **302** and the second soldering pads **303**.

The receptacle connector **201** includes an insulating body **6** and a plurality of terminal modules **7** disposed on the insulating body **6**. The insulating body **6** includes an insertion surface **61**, an insertion slot **62** extending through the insertion surface **61**, an installation space **63** communicating with the insertion slot **62**, a top wall **64** located at the top of the insertion slot **62**, and a bottom wall **65** located at the bottom of the insertion slot **62**. The insertion slot **62** is formed between the top wall **64** and the bottom wall **65** in a top-to-bottom direction. The top wall **64** is provided with a plurality of lock holes **641** communicating with the

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insertion slot **62**. The bottom wall **65** is provided with a plurality of positioning cylinders **651** protruding downwardly. The receptacle connector **201** further includes a plurality of metal grounding pieces **652** fixed on the bottom wall **65** and sleeved on the positioning cylinders **651**. The positioning cylinders **651** are adapted for being inserted into the positioning through holes **306** of the circuit board **300**. The metal grounding pieces **652** are adapted for being soldered and fixed with the ground soldering pads **307** of the circuit board **300**.

Referring to FIGS. **22** to **24**, the terminal module **7** includes a first terminal module **71** and a second terminal module **72**. During assembling, the first terminal module **71** and the second terminal module **72** are assembled into the insulating body **6** from the installation space **63**. The first terminal module **71** includes a plurality of first conductive terminals **711**, at least one first insulating block **712** and a first ground connection piece **713**. The first insulating block **712** is provided with a plurality of lock protrusions **7121** which are locked in the lock holes **641** of the insulating body **6**. In the embodiment shown in the present disclosure, the plurality of first conductive terminals **711** are insert-molded with the first insulating block **712**. The plurality of first conductive terminals **711** include a plurality of first signal terminals **S1** and a plurality of first ground terminals **G1**. Each two adjacent first signal terminals **S1** forms a first differential pair. Each side of the first differential pair is provided with one first ground terminal **G1** to improve the quality of signal transmission. The first ground connection piece **713** includes a plurality of first protrusions **7131** arranged at intervals. The plurality of first protrusions **7131** are in contact with corresponding first ground terminals **G1** to connect all the first ground terminals **G1** in series so as to improve the shielding effect. Each of the first conductive terminals **711** includes a first elastic mating portion **7111** protruding into the insertion slot **62** and a first soldering portion **7112** for being soldered and fixed with the first soldering pads **302** of the circuit board **300**.

The second terminal module **72** includes a plurality of second conductive terminals **721** and a second ground connection piece **723**. In the embodiment shown in the present disclosure, the plurality of second conductive terminals **721** are directly assembled to the insulating body **6**. The plurality of second conductive terminals **721** include a plurality of second signal terminals **S2** and a plurality of second ground terminals **G2**. Each adjacent two second signal terminals **S2** forms a second differential pair. Each side of the second differential pair is provided with one second ground terminal **G2** to improve quality of signal transmission. The second ground connection piece **723** includes a plurality of second protrusions **7231** arranged at intervals. The plurality of second protrusions **7231** are in contact with corresponding second ground terminals **G2** to connect all the second ground terminals **G2** in series so as to improve the shielding effect. Each of the second conductive terminals **721** includes a second elastic mating portion **7211** protruding into the insertion slot **62** and a second soldering portion **7212** for being soldered and fixed with the second soldering pads **303** of the circuit board **300**. The first elastic mating portions **7111** and the second elastic mating portions **7211** are respectively located on two sides (e.g., upper and lower sides) of the insertion slot **62** and face to face. The insertion slot **62** is adapted for receiving the tongue plate **22** of the plug connector **100**. The first elastic mating portions **7111** are adapted for contacting the conductive pads **221** located on one side surface (e.g., an upper surface) of the tongue plate **22**. The second elastic mating portions **7211** are

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adapted for contacting the conductive pads **221** located on the other side surface (e.g., a lower surface) of the tongue plate **22**.

The metal cage **202** includes a first insertion space **800a** and a first opening **801** located at the rear end of the first insertion space **800a** along the mating direction **M**. The receptacle connector **201** is accommodated in the metal cage **202** from the first opening **801**. The insertion slot **62** communicates with the first insertion space **800a**.

The metal cage **202** includes a first wall portion **81** (e.g., a top wall), a second wall portion **82** (e.g., a bottom wall) opposite to the first wall portion **81**, a third wall portion **83** (e.g., a right side wall) connected to the first wall portion **81**, a fourth wall portion **84** (e.g., a left side wall) opposite to the third wall portion **83**, and a fifth wall portion **85** (e.g., a rear wall) located at the rear end of the first insertion space **800a**. The first wall portion **81**, the second wall portion **82**, the third wall portion **83** and the fourth wall portion **84** are enclosed a first frame. The first insertion space **800a** extends through the first frame along a direction opposite to the mating direction **M**. In the illustrated embodiment of the present disclosure, the third wall portion **83** and the fifth wall portion **85** are provided with pressing pins **831**, **851** which are press-fitted into the conductive holes **304** of the circuit board **300**. The pressing pins **831**, **851** are respectively provided with fish-eye holes, so that the pressing pins **831**, **851** have a certain elastic deformation ability, thereby facilitating the realization of pressing the pressing pins **831**, **851** into the conductive holes **304** and maintaining reliable contact with the conductive holes **304**. The fourth wall portion **84** is provided with a plurality of hook pins **841**. The hook pins **841** are adapted for being locked in the mounting holes **305** of the circuit board **300** to increase the holding force.

In the illustrated embodiment of the present disclosure, two receptacle connectors **201** are provided and arranged side by side. The metal cage **202** includes a second insertion space **800b** arranged side by side with the first insertion space **800a**. The metal cage **202** includes a sixth wall portion **81a** (e.g., a top wall), a seventh wall portion **82a** (e.g., a bottom wall) opposite to the sixth wall portion **81a**, an eighth wall portion **83a** (e.g., a left side wall) connected to the sixth wall portion **81a**, the fourth wall portion **84** (e.g., the right side wall) opposite to the eighth wall portion **83a**, and a ninth wall portion **85a** (e.g., a rear wall) located at the rear end of the second insertion space **800b**. The sixth wall portion **81a**, the seventh wall portion **82a**, the eighth wall portion **83a**, and the fourth wall portion **84** are enclosed to form a second frame. The second insertion space **800b** extends through the second frame in the direction opposite to the mating direction **M**. In the embodiment shown in the present disclosure, the first wall portion **81** and the sixth wall portion **81a** are integrally connected and serve as a top wall of the metal cage **202**. The second wall portion **82** is integrally connected with the seventh wall portion **82a** and serves as a bottom wall of the metal cage **202**. The third wall portion **83** and the eighth wall portion **83a** are formed by bending downwardly from opposite sides of the top wall of the metal cage **202**, respectively. The fifth wall portion **85** is integrally connected with the ninth wall portion **85a** and serves as a rear wall of the metal cage **202**. The first insertion space **800a** and the second insertion space **800b** are separated by the fourth wall portion **84**. In other words, the fourth wall portion **84** is equivalent to a partition wall to divide the metal cage **202** into the first insertion space **800a** and the second insertion space **800b** which are located adjacent to each other.

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Specifically, the metal cage **202** includes a first metal shell **8a** and a second metal shell **8b** assembled and fixed with the first metal shell **8a**. The first metal shell **8a** includes the first wall portion **81**, the sixth wall portion **81a**, the third wall portion **83**, and the eighth wall portion **83a**. The second metal shell **8b** includes the second wall portion **82** and the seventh wall portion **82a**. Since the structures forming the first insertion space **800a** and the second insertion space **800b** are similar, only the first insertion space **800a** is taken as an example for detailed description below.

Referring to FIGS. **25** to **33**, in the illustrated embodiment of the present disclosure, the second wall portion **82** includes a retaining portion **821** and a dovetail protrusion **822** located at the front end of the retaining portion **821**. The retaining portion **821** includes a first locking hole **8211**, a first buckle tab **8212** protruding into the first locking hole **8211** along an assembling direction (e.g., a bottom-to-top direction), and a first abutting surface **8213** opposite to the first buckle tab **8212**. In the embodiment shown in the present disclosure, the retaining portion **821** is formed by bending the second wall portion **82**.

The third wall portion **83** includes a first locking protrusion **8311** held in the first locking hole **8211** and a dovetail groove **832** for holding the dovetail protrusion **822**. The first locking protrusion **8311** includes a first retaining groove **8312** mated with the first buckle tab **8212** and a first limiting surface **8313** exposed in the first retaining groove **8312**. The first abutting surface **8213** cooperates with the first locking protrusion **8311** to restrict the second metal shell **8b** from being separated from the first metal shell **8a** in a direction opposite to the assembling direction. The first limiting surface **8313** cooperates with the first buckle tab **8212** to limit the excessive displacement of the second metal shell **8b** relative to the first metal shell **8a** along the assembling direction. It is understandable to those skilled in the art that the term “excessive displacement” used in the present disclosure includes but is not limited to the following scenarios:

First scenario: When the second metal shell **8b** is assembled to the first metal shell **8a** along the assembling direction, the first limiting surface **8313** cooperates with the first buckle tab **8212**, the second metal shell **8b** can be prevented from being over-assembled relative to the first metal shell **8a** along the assembling direction.

Second scenario: When the second metal shell **8b** has been assembled to the first metal shell **8a** and is subjected to an external force, through the cooperation between the first limiting surface **8313** and the first buckle tab **8212**, the second metal shell **8b** can be prevented from further moving toward the first metal shell **8a**.

In the embodiment shown in the present disclosure, the first locking protrusion **8311** includes a second abutting surface **8311b** located at an end of the first locking protrusion **8311** along the assembling direction. The second abutting surface **8311b** is configured to abut against the first abutting surface **8213** to prevent the first locking protrusion **8311** from disengaging from the first locking hole **8211** in the direction opposite to the assembling direction.

In the illustrated embodiment of the present disclosure, the first buckle tab **8212** includes a first pressing surface **8212a** located at an end of the first buckle tab **8212** along the assembling direction. The first limiting surface **8313** is configured to press against the first pressing surface **8212a**, so as to prevent the second metal shell **8b** from being excessively displaced relative to the first metal shell **8a** along the assembling direction.

In the embodiment shown in the present disclosure, the first locking protrusion **8311** includes a first locking surface

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8311c and a second locking surface **8311d** spaced along the front-rear direction. The retaining portion **821** includes a first matching surface **8214** and a second matching surface **8215** exposed in the first locking hole **8211** along the front-rear direction. The first mating surface **8214** is configured to abut against the first locking surface **8311c** to prevent the first locking protrusion **8311** from moving forwardly. The second mating surface **8215** is configured to abut against the second locking surface **8311d** to prevent the first locking protrusion **8311** from moving backwardly.

In the embodiment shown in the present disclosure, the first locking hole **8211** and the first retaining groove **8312** are surrounding types, so as not to significantly reduce the corresponding structural strength of the first metal shell **8a** and the second metal shell **8b**. In the embodiment shown in the present disclosure, the first locking protrusion **8311** is a stamping protrusion formed by stamping outward from the third wall portion **83**. The first locking protrusion **8311** includes an inclined guide surface **8311a**. The inclined guide surface **8311a** is configured to guide the retaining portion **821** and abut against the retaining portion **821** when the second metal shell **8b** is assembled to the first metal shell **8a**. After the second metal shell **8b** and the first metal shell **8a** are assembled, the first locking protrusions **8311** are tightly held in first locking holes **8211** along the front-rear direction perpendicular to the assembling direction, so as to realize the position limit in the front-rear direction and prevent loosening of the first metal shell **8a** and the second metal shell **8b** in the front-rear direction.

The third wall portion **83** and/or the fourth wall portion **84** is provided with a first guide protrusion **89** which protrudes into the first insertion space **800a**. The first guide protrusion **89** is configured to be inserted into the slot **13** of the plug connector **100**. In the embodiment shown in the present disclosure, both the third wall portion **83** and the fourth wall portion **84** are provided with the first guide protrusions **89** which protrude into the first insertion space **800a**. The first guide protrusion **89** of the third wall portion **83** and the first guide protrusion **89** of the fourth wall portion **84** are respectively inserted into the slots **13** located on two sides of the plug connector **100** in order to improve mating stability. In the illustrated embodiment of the present disclosure, the first guide protrusions **89** are punched from the third wall portion **83** and the fourth wall portion **84**, respectively.

The receptacle connector assembly **200** includes a first chamber **802** between the first wall portion **81** and the insulating body **6**. The third wall portion **83** and/or the fourth wall portion **84** is provided with a second guide protrusion **86** which protrudes into the first chamber **802**. The first guide protrusion **89** is configured to guide the extension plate **113** of the plug connector **100**. The second guide protrusion **86** is configured to support the extension plate **113** of the plug connector **100**. In the embodiment shown in the present disclosure, the heights of the first guide protrusions **89** and the second guide protrusions **86** are different, but both extend in the horizontal direction. The first guide protrusions **89** and the second guide protrusions **86** are both perpendicular to the third wall portion **83** and the fourth wall portion **84**. The first guide protrusion **89** and the second guide protrusion **86** are located on the front and rear sides of the receptacle connector **201**, respectively.

When the plug connector **100** is inserted into the receptacle connector assembly **200**, the first guide protrusion **89** supports and guides the extension plate **113**. When the plug connector **100** is inserted into place, the tongue plate **22** is inserted into the insertion slot **62**, the first elastic mating portions **7111** and the second elastic mating portions **7211**

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abut against the conductive pads **221**, and the first guide protrusion **89** is inserted into the slot **13** to improve mating stability. During the insertion process of the plug connector **100**, the first guide protrusion **89** first abuts against the lower surface **1133** of the extension plate **113**; then the first guide protrusion **89** abuts against the transition inclined surface **1134**; and finally the first guide protrusion **89** abuts against the slot top surface **131** of the housing **1**. Since the slot top surface **131** is lower than the lower surface **1133** of the extension plate **113** along the thickness direction T-T, when the plug connector **100** is initially inserted (that is, when the first guide protrusion **89** abuts against the lower surface **1133** of the extension plate **113**), the reaction force exerted on the plug connector **100** is small, so that the insertion and extraction force of the plug connector **100** in the early stage of insertion is small. When the plug connector **100** is inserted into the later stage (that is, after the first guide protrusion **89** crosses the transition inclined surface **1134** and abuts against the slot top surface **131** of the housing **1**), the reaction force exerted on the plug connector **100** is larger, which is beneficial to stabilize the plug connector **100** in the mating position. It is understandable to those skilled in the art that the change in the plugging and unplugging feel of the plug connector **100** is helpful for the user to determine the insertion position of the plug connector **100**.

The metal cage **202** includes a plurality of grounding springs **87** assembled to the first metal shell **8a** and/or the second metal shell **8b**. In the embodiment shown in the present disclosure, the plurality of grounding springs **87** include a first grounding spring **87a**, a second grounding spring **87b**, a third grounding spring **87c**, and a fourth grounding spring **87d**. The first grounding spring **87a** is fixed to the first wall portion **81**. The second grounding spring **87b** is fixed to the second wall portion **82**. The third grounding spring **87c** is fixed to the third wall portion **83**. The fourth grounding spring **87d** is fixed to the fourth wall portion **84**. Each grounding spring **87** includes a plurality of first elastic arms **871** protruding into the first insertion space **800a**, a plurality of second elastic arms **872** located outside the first insertion space **800a**, and a U-shaped connecting portion **873** connecting the plurality of first elastic arms **871** and the plurality of second elastic arms **872**. The first grounding spring **87a** and the second grounding spring **87b** are symmetrically arranged on the upper and lower sides of the first insertion space **800a**. Referring to FIG. **25**, the first metal shell **8a** and/or the second metal shell **8b** are provided with positioning protrusions **88**. The U-shaped connecting portion **873** of the first grounding spring **87a** and/or the second grounding spring **87b** is provided with a plurality of positioning holes **8731** mated with the positioning protrusions **88** to facilitate assembly and positioning. In the embodiment shown in the present disclosure, the first metal shell **8a** and the second metal shell **8b** are both provided with the positioning protrusions **88**. The U-shaped connecting portions **873** of the first grounding spring **87a** and the second grounding spring **87b** are both provided with the positioning holes **8731** mated with the positioning protrusions **88**.

Referring FIGS. **25** to **27** and FIG. **33**, the plurality of first elastic arms **871** of the first grounding spring **87a** and the second grounding spring **87b** include a plurality of first arc-shaped arms **8711** and a plurality of first straight arms **8712**. Each first straight arm **8712** includes a first flip portion **8712a**. The first wall portion **81** and the second wall portion **82** include first retaining holes **203**. The first flip portions **8712a** pass through the first retaining holes **203** to be fixed on the metal cage **202**. In the illustrated embodiment of the present disclosure, the first straight arms **8712** abut against

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an inner side of the corresponding first wall portion **81** and an inner side of the second wall portion **82**. The first arc-shaped arm **8711** protrudes into the first insertion space **800a** to abut against the plug connector **100** so as to achieve grounding purpose and increase the insertion force. In the embodiment shown in the present disclosure, by disposing the first straight arms **8712**, when the plug connector **100** is inserted into the first insertion space **800a**, the first straight arms **8712** can be prevented from being excessively pressed to cause the first straight arms **8712** to be loosened.

The third grounding spring **87c** includes a plurality of second arc-shaped arms **8715** and at least one second straight arm **8716**. The at least one second straight arm **8716** includes a second flip portion **8716a**. The third wall portion **83** includes a second retaining hole **205**. The second flip portion **8716a** passes through the second locking hole **205** to be fixed on the metal cage **202**. In the illustrated embodiment of the present disclosure, the second straight arm **8716** abuts against an inner side of the third wall portion **83**. The second arc-shaped arms **8715** protrude into the first insertion space **800a** to abut against the plug connector **100** so as to achieve grounding purpose and increase the insertion force. In the embodiment shown in the present disclosure, by disposing the second straight arm **8716**, when the plug connector **100** is inserted into the first insertion space **800a**, the second straight arm **8716** can be prevented from being excessively pressed to cause the second straight arm **8716** to be loosened.

The metal cage **202** includes a locking groove **204** which is matched with the locking latch **411** of the plug connector **100**. The first elastic arm **871** is provided with an escape groove **8713** corresponding to the locking groove **204**. The at least one grounding spring **87** includes a soldering point **8714** located adjacent to the locking groove **204** and located at a front end of the locking groove **204** in the mating direction M of the plug connector **100**. It is understandable to those skilled in the art that at the position where the locking latch **411** mates with the locking groove **204**, when the plug connector **100** is unlocked and pulled out, the metal cage **202** and the first elastic arms **871** of the first grounding spring **87a** are easily pulled. By arranging the soldering point **8714** adjacent to the locking groove **204** in the present disclosure, the structure can be strengthened without excessively increasing the manufacturing cost. Of course, in other embodiments of the present disclosure, the positioning protrusions **88** are matched with the positioning holes **8731** of the U-shaped connecting portion **873**, and by fixing the first flip portion **8712a** to the metal cage **202**, the first elastic arm **871** of the present disclosure can even omit the soldering step, thereby reducing the difficulty of manufacturing.

Compared with the prior art, the present disclosure provides a first guide protrusion **89** protruding into the first insertion space **800a**. When the plug connector **100** is inserted into the receptacle connector assembly **200**, the first guide protrusion **89** supports and guides the extension plate **113** of the plug connector **100**. When the plug connector **100** is inserted in place, the first guide protrusion **89** is inserted into the slot **13** of the plug connector **100**, thereby improving the mating stability. In addition, in the present disclosure, by providing the first locking hole **8211** and the first buckle tab **8212** on the retaining portion **821**, by providing the first locking protrusion **8311** and the first retaining groove **8312** on the at least one wall portion, by fixing the first locking protrusion **8311** in the first locking hole **8211**, and by receiving the first buckle tab **8212** in the first retaining groove **8312**, the first metal shell **8a** and the second metal shell **8b** are assembled and fixed, thereby reducing the risk

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of loosening of the first metal shell **8a** and the second metal shell **8b** and improving the structural reliability.

It should be noted that, in the present disclosure, a general concept of the receptacle connector assembly **200** is an electrical connector. In the embodiment shown in the present disclosure, the metal cage **202** of the electrical connector mainly plays a role of at least partially shielding the insulating body **6**, while the connection relationship between the metal cage **202** and the insulating body **6** is not significantly important. It is understandable to those skilled in the art that, in other embodiments, in addition to the function of at least partially shielding the insulating body **6**, the metal cage **202** of the electrical connector is also closely connected with the insulating body **6**. For example, the insulating body **6** is installed and fixed in the metal cage **202**. In addition, although the electrical connector is the receptacle connector assembly **200** in the illustrated embodiment of the present disclosure. In other embodiments, the electrical connector may also be a plug connector or any electrical connector with the metal cage **202**.

The above embodiments are only used to illustrate the present disclosure and not to limit the technical solutions described in the present disclosure. The understanding of this specification should be based on those skilled in the art. Descriptions of directions, although they have been described in detail in the above-mentioned embodiments of the present disclosure, those skilled in the art should understand that modifications or equivalent substitutions can still be made to the application, and all technical solutions and improvements that do not depart from the spirit and scope of the application should be covered by the claims of the application.

What is claimed is:

1. A connector assembly, comprising: a receptacle connector assembly and a plug connector;

the receptacle connector assembly comprising:

a receptacle connector, the receptacle connector comprising an insulating body and a plurality of conductive terminals disposed on the insulating body; the insulating body comprising an insertion surface and an insertion slot extending through the insertion surface; and

a metal cage, the metal cage at least partially shielding the receptacle connector;

the metal cage defining a first insertion space in communication with the insertion slot;

the plug connector comprising:

a housing, the housing comprising a mating surface; and

a tongue plate on which a plurality of conductive pads are provided;

wherein the housing comprises an extension plate extending along a mating direction of the plug connector; the tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction; the plug connector is provided with the extension plate which is located only on one side of the tongue plate; the extension plate extends and protrudes beyond the tongue plate along the mating direction;

wherein the housing comprises a first housing and a second housing, the first housing comprises a first base and a first extension portion, the second housing comprises a second base and a second extension portion, and the extension plate extends from the first extension portion along the mating direction;

wherein the first extension portion comprises a first surface; the second extension portion comprises a second

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surface; the first surface and the second surface are disposed oppositely along a thickness direction of the plug connector;

each of two sides of the plug connector further defines a slot extending along the mating direction and extending through the mating surface.

2. The connector assembly according to claim 1, wherein a length of the extension plate protruding beyond the mating surface along the mating direction is $L1$, and a length of the tongue plate protruding beyond the mating surface along the mating direction is $L2$, where $L1 \geq 2 * L2$.

3. The connector assembly according to claim 1, wherein the extension plate is provided with a guide portion located at an end of the extension plate along the mating direction; an upper surface of the guide portion and/or a lower surface of the guide portion is a guide inclined surface.

4. The connector assembly according to claim 1, wherein the first extension portion extends from the first base, the second extension portion extends from the second base, the first base corresponds to the second base in the thickness direction of the plug connector, and the first extension portion corresponds to the second extension portion in the thickness direction.

5. The connector assembly according to claim 1, wherein a distance between the slot and the first surface in the thickness direction is smaller than a distance between the slot and the second surface.

6. The connector assembly according to claim 1, wherein the extension plate comprises an upper surface and a lower surface opposite to the upper surface of the extension plate; a distance between the upper surface of the tongue plate and the lower surface of the extension plate along the thickness direction is greater than a distance between the lower surface of the tongue plate and the second surface of the second extension portion along the thickness direction.

7. The connector assembly according to claim 6, wherein the upper surface of the extension plate is coplanar with the first surface of the first extension portion;

the housing comprises a slot top surface downwardly exposed in the slot; the slot top surface is lower than the lower surface of the extension plate along the thickness direction;

the housing further comprises a transition inclined surface connecting the lower surface of the extension plate and the slot top surface.

8. The connector assembly according to claim 1, wherein the first extension portion comprises two convex portions protruding along the thickness direction and located on two sides of the first extension portion; each convex portion extends along the mating direction; each slot is provided on a corresponding convex portion.

9. The connector assembly according to claim 8, wherein the convex portion comprises a first end surface; the second extension portion comprises a second end surface; the first end surface and the second end surface are aligned with each other in the thickness direction to jointly form the mating surface.

10. The connector assembly according to claim 8, wherein a side of the convex portion comprises a locking portion; a side of the second extension portion defines a locking slot; the locking portion is fixed in the locking slot to prevent the first housing and the second housing from being separated from each other in the thickness direction.

11. The connector assembly according to claim 1, wherein the metal cage comprises a first opening located at an end of

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the first insertion space along the mating direction; the receptacle connector is received in the metal cage from the first opening;

the extension plate extends and protrudes beyond the slot along the mating direction;

the metal cage comprises a first wall portion, a second wall portion opposite to the first wall portion, a third wall portion connected to the first wall portion and a fourth wall portion opposite the third wall portion; the first wall portion, the second wall portion, the third wall portion and the fourth wall portion jointly form a first frame; the first insertion space extends through the first frame along a direction opposite to the mating direction; the third wall portion and/or the fourth wall portion comprise a first guide protrusion protruding into the first insertion space and a second guide protrusion protruding into the first opening;

when the plug connector is initially inserted into the receptacle connector assembly, the first guide protrusion is configured to guide the extension plate of the plug connector; and

when the plug connector is fully inserted into the receptacle connector assembly, the first guide protrusion is configured to be inserted into the slot of the plug connector, and the second guide protrusion is configured to support the extension plate of the plug connector.

12. A plug connector, comprising:

a housing, the housing comprising a mating surface;

a tongue plate on which a plurality of conductive pads are provided; and

a plurality of cables electrically connected to the plurality of conductive pads;

wherein the housing comprises an extension plate extending along a mating direction of the plug connector; the tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction; the plug connector is provided with the extension plate which is located only on one side of the tongue plate; the extension plate extends and protrudes beyond the tongue plate along the mating direction;

wherein the housing comprises a first housing and a second housing, the first housing comprises a first base and a first extension portion, the second housing comprises a second base and a second extension portion, and the extension plate extends from the first extension portion along the mating direction;

wherein the first extension portion comprises a first surface; the second extension portion comprises a second surface; the first surface and the second surface are disposed oppositely along a thickness direction of the plug connector;

each of two sides of the plug connector further defines a slot extending along the mating direction and extending through the mating surface.

13. The plug connector according to claim 12, wherein a length of the extension plate protruding beyond the mating surface along the mating direction is L1, and a length of the tongue plate protruding beyond the mating surface along the mating direction is L2, where $L1 \geq 2 * L2$.

14. The plug connector according to claim 12, wherein the extension plate is provided with a guide portion located at an end of the extension plate along the mating direction; an upper surface of the guide portion and/or a lower surface of the guide portion is a guide inclined surface.

15. The plug connector according to claim 12, wherein the first extension portion extends from the first base, the second

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extension portion extends from the second base, the first base corresponds to the second base in the thickness direction of the plug connector, and the first extension portion corresponds to the second extension portion in the thickness direction.

16. The plug connector according to claim 12, wherein a distance between the slot and the first surface in the thickness direction is smaller than a distance between the slot and the second surface.

17. The plug connector according to claim 12, wherein the extension plate comprises an upper surface and a lower surface opposite to the upper surface of the extension plate; a distance between the upper surface of the tongue plate and the lower surface of the extension plate along the thickness direction is greater than a distance between the lower surface of the tongue plate and the second surface of the second extension portion along the thickness direction.

18. The plug connector according to claim 17, wherein the upper surface of the extension plate is coplanar with the first surface of the first extension portion;

the housing comprises a slot top surface downwardly exposed in the slot; the slot top surface is lower than the lower surface of the extension plate along the thickness direction;

the housing further comprises a transition inclined surface connecting the lower surface of the extension plate and the slot top surface.

19. The plug connector according to claim 12, wherein the plug connector is configured to mate with a receptacle connector assembly; the receptacle connector assembly comprising:

a receptacle connector, the receptacle connector comprising an insulating body and a plurality of conductive terminals disposed on the insulating body; the insulating body comprising an insertion surface and an insertion slot extending through the insertion surface; and a metal cage, the metal cage at least partially shielding the receptacle connector; the metal cage defining a first insertion space in communication with the insertion slot;

wherein the metal cage comprises a first opening located at an end of the first insertion space along the mating direction; the receptacle connector is received in the metal cage from the first opening;

the plug connector defines a slot; the extension plate extends and protrudes beyond the slot along the mating direction;

the metal cage comprises a first wall portion, a second wall portion opposite to the first wall portion, a third wall portion connected to the first wall portion and a fourth wall portion opposite the third wall portion; the first wall portion, the second wall portion, the third wall portion and the fourth wall portion jointly form a first frame; the first insertion space extends through the first frame along a direction opposite to the mating direction; the third wall portion and/or the fourth wall portion comprise a first guide protrusion protruding into the first insertion space and a second guide protrusion protruding into the first opening;

when the plug connector is initially inserted into the receptacle connector assembly, the first guide protrusion is configured to guide the extension plate of the plug connector; and

when the plug connector is fully inserted into the receptacle connector assembly, the first guide protrusion is configured to be inserted into the slot of the plug

connector, and the second guide protrusion is configured to support the extension plate of the plug connector.

20. A plug connector, comprising:
- a housing, the housing comprising a mating surface; 5
 - a built-in circuit board partially located in the housing, the built-in circuit board including a tongue plate on which a plurality of conductive pads are provided; and
 - a plurality of cables electrically connected to the built-in circuit board; 10
- wherein the housing comprises an extension plate extending along a mating direction of the plug connector; the tongue plate and the extension plate both extend and protrude beyond the mating surface along the mating direction; the extension plate extends and protrudes 15 beyond the tongue plate along the mating direction;
- wherein a length of the extension plate protruding beyond the mating surface along the mating direction is L1, and a length of the tongue plate protruding beyond the mating surface along the mating direction is L2, where 20 $L1 \geq 2 * L2$;
- wherein each of two sides of the plug connector further defines a slot extending along the mating direction and extending through the mating surface; the slot is configured to receive a first guide protrusion of a metal 25 cage into which the plug connector is at least partially inserted.

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